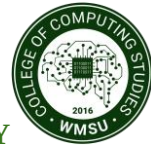




Republic of The Philippines
Western Mindanao State University
COLLEGE OF COMPUTING STUDIES
DEPARTMENT OF INFORMATION TECHNOLOGY



NUTRI APP: NUTRITION LEARNING APPLICATION USING AUGMENTED REALITY FOR KIDS

A CAPSTONE PROJECT

Presented to the Faculty of the
College of Computing Studies
Western Mindanao State University

In Partial Fulfillment
of the Requirements for the Degree
Bachelor of Science in Information Technology

By

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EXECUTIVE SUMMARY

The use of traditional way of learning has always been part of the education system of the Department of Education. The use of textbooks to impart knowledge based on a relevant curricular topic for the learners. The traditional teaching approach has been used for a long time in the education system of the Philippines, where students are just expected to sit quietly in their seats and listen to the professors' didactic manner, which makes them passive participants in class. Another phrase for how teachers impart knowledge and understanding of concepts and ideals to students is "spoon feeding."

The project aims to assist the learners get the most out of how they use technology. The users' attention is piqued, and their enthusiasm is increased by learning the nutritional details while trying to earn rewards and having fun with the integrated augmented reality. They get more involved in the learning process as a result. Children can use technology in a wide range of contexts. Using a phone or other technology is one of the many activities' kids engage in on a daily basis. The functionality of utilizing accessible phones will be more beneficial and purposeful with the help of this application because the kids will learn while having fun. The researchers discovered that the selected learners from Zamboanga City who took the survey are satisfied with the overall performance of the application and they would recommend it to their friends and classmates. The use of marker-based augmented reality learning to impart nutritional concept will boost attentiveness and interaction has also been proven on this project. Therefore, making this application an alternative learning tool for nutrition learning application.

Keywords: Nutri-App, Nutritional Concepts, Marker-based Augmented Reality, Zamboanga City.

DEDICATION

We, the researchers, dedicate this to our family. Who supported and believed in us, who has been our source of inspiration and motivation to overcome the challenges we faced throughout our college journey. The love and support from our families are the primary backbone of our victories. To our friends, we could not have achieved this without your moral support, you helped us redirect our path at times when we were at our lows. To our special ones, the motivation you gave and support you've given was unparalleled. Thank you for existing and for being there. To our adviser, we are so grateful for your support, without your guidance and knowledge this project wouldn't be possible. Finally, thank you to the Lord, for the perseverance, courage, skill sets and focused mind. We are eternally grateful for these blessings.

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CHAPTER 1

INTRODUCTION

1.1 Project Context

In Central Luzon, in the year 2020-2021 the Department of Education Region III's Health and Nutrition Unit has started health and nutrition programs for public kindergarten and elementary school students as part of its COVID-19 prevention efforts.

The DepEd School Health and Nutrition Program continually encourage students to adopt healthy habits, even when they are at home. The distribution of learners' hygiene kits and the school-based feeding program are two steps in the direction of better learning leading to better health, according to Dra. Gladys Lourdes Bengco, Medical Officer IV for DepEd Region III.

Children in elementary school learn about nutrition in the classroom using conventional teaching methods. Verbal explanation and visual aids. On the use of the technological side, the learning modality of learning comes from educational videos like YouTube for kids and learning applications for kids through online learning. These applications provide the learning process through learning visual materials and interactive listening.

The different educational methods were created as part of the Basic Education - Learning Continuity Plan, which addresses crucial criteria for education during and even after the COVID-19 period. The usage of learning modalities includes Online Distance Learning which features the teacher as facilitator, engaging learners' active participation through the use of various technologies accessed through the internet while they are geographically remote from each other during instruction. Blended Learning refers to a learning delivery that combines face-to-face with any or a mix of online distance learning.

Technology has revolutionized the way we learn and educate ourselves. Especially during the pandemic, technology has become the primary medium for education. The use of technology has ushered in a new era of teaching and learning. By utilizing various applications, technology can enhance the learning of nutritional concept.

This project provides a learning application for children aged 6-9 years old, with an interesting feature of technology, as a kid with endless curiosity, this application will increase the engagement and motivation on learning nutritional concepts through an interactive mobile learning application. The application will recognize 2D images using marker-based augmented reality, the scanned image will appear on the screen with labeled nutrition facts and what type of food this is based on the food pyramid. The integration of augmented reality in the application will increase the interest and motivation in learning nutritional concepts. This application will also have mini-games that will have a reward-based module for the users.

1.2 Purpose and Description

Traditional teaching was the long-established teaching style in the Philippine education system wherein the students are passive in terms of participating in class, due to only having to sit quietly in their places and listen to the teachers' didactic method. Teachers transmit knowledge and understanding of ideas and values to the student another term is "spoon feeding". When the classroom is teacher-centered, the classroom becomes orderly and the teacher can control the flow of the class easily. (Chi-ki, 2010) This method does not promote curiosity as long as the pupil is disinterested, they will not participate unless the teacher will call them. On the other hand, teacher-centered learning positions the teacher as the expert in charge of imparting knowledge to his or her students via lectures or direct instruction. With the extended help of modernized teaching methods, such as imparting

augmented reality and mini-games to a particular learning concept. This will be of good benefit to the traditional teaching style of primary teachers, it will increase the focus and engagement in learning nutritional concepts for the students.

In teacher-centered learning, the teacher is considered the expert and is responsible for imparting knowledge to students through lectures or direct instruction. However, modern teaching methods like augmented reality and mini-games can be incorporated to supplement traditional teaching methods. This can greatly benefit primary teachers by increasing student engagement and focus when learning about nutritional concepts. By utilizing technology in the classroom, teachers can create a more dynamic and interactive learning environment that caters to the different learning styles and interests of students.

This can result in a more effective and enjoyable learning experience for everyone involved. One of the children's many activities in their daily routine is using mobile phones or interacting with technology.

This application will enhance the usefulness and effectiveness of accessible phones for children's learning by making it fun and engaging. By incorporating augmented reality into the nutrition learning process, children can stay focused and interact more, even during their leisure time.

1.3 Project Objectives

a. General Objective

The general objective of this project is to provide an alternative learning tool for learning nutritional foods through interactive engagements.

b. Specific Objectives

1. Use marker-based augmented reality to impart nutritional concepts such as Glow, Go, and Grow Foods.

2. Utilize multimedia tools through games and audio/voice-over to increase interactivity.
3. Develop the application using Unity 2D and Vuforia Engine.
4. Test the usability of the learning app through end-user evaluation.

1.4 Scopes and Limitations

The study is directed at children aged six to nine years old. The concept and content of the lesson from the application are based on the learning materials from the Department of Education, Zamboanga City. To start the learning process using the application, users are presented with two options in the main menu: lessons and challenges. The lessons section includes Go foods, Grow foods, and Glow foods, which incorporate augmented reality features. Users must scan the appropriate 2D image for the app to project the corresponding augmented reality image. The app provides voice-over feedback that offers relevant nutrition facts related to the scanned 2D image. Additionally, the application includes a food pyramid and allows users to access descriptions for each food category.

The challenges section features the relaying of learning processes through fun and interactive games that give rewards to the users after completing tasks. A food pyramid game that will let the user drag and drop food, vegetables, or fruit that falls into their corresponding category. Another game is a similar concept of a food pyramid where-in the users are tasked to drag and drop a variety of healthy food into its appropriate basket. On the main menu, the users can access the Google drive link of the downloadable pdf file for the scanning of the marker-based augmented reality.

And lastly, a game quiz that asks the users five questions per food category based on the voice-over feedback of the categories glow, grow, and glow foods.

The application's compatibility is with the android operating system version 9 or the latest versions available. The application has affirmation features that give feedback on the actions of the user in accordance with their right or wrong answers, clicks, and inaccurate navigation while browsing. The application was developed in unity 2d using the software development kit vuforia engine, for rendering marker-based augmented reality.

1.5 Significance of the Study

The theory of intrinsic motivation was applied to explain why people are motivated to learn. The Instructional Materials Motivation Survey was used to develop the research instrument as the attention, relevance, confidence, and satisfaction (ARCS) model served as the framework for understanding the effects of augmented reality on student motivation. The study looked at the variations in students' motivation for learning both before and after using the augmented reality mobile application. The augmented reality mobile application was utilized by 78 participants, who also answered pre- and post-use surveys. The results demonstrate that using mobile augmented reality applications enhanced students' enthusiasm to learn. These findings were significant and showed an increase in the motivational factors of attention, satisfaction, and confidence. (Khan, 2019)

The beneficiaries of this project are:

1. Children aged 6-9 years old - who will learn about appropriate nutritional concepts in a fun and engaging way using augmented reality.
2. Researchers and developers - who can use this project as a conceptual foundation for future integration of modern technology in learning, particularly in increasing engagement and motivation.
3. Public schools - who can benefit from this project as it provides supported learning methods that can increase the productivity of teachers.

The project aims to assist the children to make the most out of their technology utilization. Learning the nutritional facts while aiming to gain rewards and having fun with the integrated augmented reality piques the interest and boosts the users' enthusiasm. This helps them engage more while they get to absorb educational information.

The teachers of the users can use this application as an alternative way of learning nutritional concepts through augmented reality and games.

The learning materials and context of the application will help the learners aged from six to nine years old have the opportunity to learn nutritional facts while having fun. The entire process will and can be a guide for the kids to be knowledgeable about dietary needs and what foods are healthy.

1.6 Definition of Terms

Attention, Relevance, Confidence and Satisfaction Model of Motivation - is a motivational design process that includes a synthesis of motivational concepts and theories that are clustered into four categories: attention (A), relevance (R), confidence (C), and satisfaction (S). This model is particularly important for e-learning, since motivating learners in an online course more difficult than in face-to-face courses.

Augmented Reality - A technology that superimposes a computer-generated image on a user's view of the natural world, thus providing a composite view.

Didactic Method of Teaching - Didactic teaching is a type of instruction that emphasizes teachers instructing students. Teachers who employ didactic methods frequently design organized lessons with a strong lecture component. Depending on the needs of their pupils, their own preferences, and other circumstances, teachers might utilize a variety of teaching techniques.

End-user Evaluation - is a process of assessing the effectiveness, usability, and overall quality of a product or service from the perspective of the end-users or customers. The evaluation involves gathering feedback and opinions from users about their experience using the product or service, including its features, design, performance, and usability.

Interactive Learning - is an educational approach that emphasizes active participation and engagement of students in the learning process. It involves creating opportunities for students to interact with the learning materials, such as through discussion, problem-solving, experiments, simulations, and games.

Marker-based Augmented Reality - is a type of augmented reality that uses specific markers, which are often 2D barcodes or images with unique patterns, as reference points to place and orient virtual objects in the real world.

Teacher-Centered Approach - The teacher-centered model places the instructor in the role of the subject-matter expert in charge of educating his or her students through lectures or hands-on instruction.

Unity 2d - Unity is a cross-platform game engine developed by Unity Technologies, first announced and released in June 2005 at Apple Inc.'s Worldwide Developers Conference as a Mac OS X-exclusive game engine.

Vuforia Engine - is a software development kit for creating Augmented Reality apps. Developers can easily add advanced computer vision functionality to any application, allowing it to recognize images and objects, and interact with spaces in the real world.

CHAPTER 2

REVIEW OF RELATED LITERATURE

2.1 Introduction

This study reviews the benefits of integrating technology in education. The advantageous impact of augmented reality (AR) in teaching are examined in these studies. It also determines the user's adaptability to the use of marker-based augmented reality. The researchers developed a nutritional learning mobile application with the integration of augmented reality and interactive games.

2.2 Learning with the Integration of Technology

It is evident that using mobile-based augmented reality learning material for children is effective and increases the interactivity of imparting augmented reality into learning. This paper outlines 23 usability concepts that will be used to create an interactive mobile-based augmented reality application for learning basic English. Qualitative and quantitative analyses were used to determine the usability and acceptability of the proposed application in the eyes of users. The planned application's use is supported by an analysis of students' behavior and interviews with teachers. A study on Evaluating Usability of Mobile-Based Augmented Reality Learning Environments for Early Childhood. (Tuli N, Mantri A, 2021) Learning tool for kids on the android platform is effective as this study suggests that if a mobile application is attractive, the children will have their undivided attention which will result in an intact learning method. This paper describes an Android-based educational application called "Homy Kid Lesson." The purpose of this "Homy Kid Lesson" is to assist children in learning numbers, letters, and animal names, and performing simple calculations using an appealing interface. Data collection, design, and implementation are all examples of research methods.

The research literature is used to develop the app, which is then compared to similar apps, interviews, and questionnaires. (Gulam, Salman A, Kanigoro B, 2020)

An Educational Mobile City Learning Application for Kids. With the advancement of technology, mobile devices have begun to pervade many aspects of life. One of these is the entertainment industry, because of these devices' visual and interactive interfaces.

The goal of this project is to create a mobile application that allows youngsters to enjoy learning about their country's cities. This program will ensure that students recognize history, geography, and historical events by combining games and education. (Ocan M, Goz F, 2017)

User interface design guidelines for children's mobile learning applications. It is evident in this study that with an interactive and appealing graphical user interface, the learning experience for children will be enhanced. Mobile learning applications that come with a good and effective user interface allow users to be aware and mindful of the subject matter. This enables users to fully optimize the applications for learning purposes. (Latiff H, Razali R, Ismail F, 2019)

2.3 Augmented Reality in Learning Applications

Learning strategies using augmented reality technology in education: Meta-analysis. This study suggests the efficacy of strategies in learning with the use of augmented reality. Users can interact with virtual things that are merged into the actual world and appear in the same place in real-time using Augmented Reality technology. The goal of this meta-analysis is to identify the different types of Augmented Reality learning methodologies that have been used. (Hanid M, Mohamad Said M, Yahaya N, 2020) Virtual and augmented reality in education presents a survey of the literature on virtual and augmented reality, as well as an examination of integrated models for implementing educational processes.

The strategical use of augmented reality in approaching an alternative way in teaching shows a promising result. And with a touch of emerging technology the learning process will be more interesting and timelier. (Guidoniene D, Rutkauskiene D, 2019) Resulting in the provision of a model for the development of integrated learning objects based on a VR and AR approach, where VR and AR objects could be integrated into other educational content as learning objects by creating integrated learning programs, learning scenarios, or other learning objects.

Virtual Reality and Augmented Reality in education. This review shows how and why Virtual Reality or Augmented Reality tools and applications contribute to the learning of new knowledge and skills. Virtual Reality (VR) and Augmented Reality (AR) technologies and solutions are paving the way for new learning and education opportunities.

Using a core set of literature reviews, the team explored how and why VR/AR tools and applications contribute to the learning of new knowledge and skills, as well as the research and development of VR/AR at Utrecht University. (Fajardo Tovar D, Jonker V, Hürst W, 2020) A review on Augmented Reality for Education suggests the extensive effects of integrating augmented reality education. The incorporation of information in digital format that incorporates live footage of a user's real-time environment is known as Augmented Reality, or simply AR.

Various universities are currently embracing Augmented Reality as well. The use of technology in the education industry can lead to the creation of a smart campus. This is essential to this project, the integration of augmented reality in giving a new perspective in learning, as it emphasizes the promising long-term effect of using modern technology in education.

2.4 Augmented Reality in Education and its Advantageous Effects

Augmented Reality and Virtual Reality in education. “Myth or reality?” in this study shows the effectiveness of integrating virtual reality in education for 21st-century learners.

AR and VR have become more viable and desirable in many sectors, including education, because of recent technology advancements and the spread of affordable gear and software; they have been relaunched with fresh promises previously imagined. Because of the nature of AR and VR, new teaching and learning methods that better address the demands of 21st-century learners are on the horizon. This explains the reasons behind the recent rise of AR and VR, as well as why their widespread acceptance in education will be a reality soon. (Elmqaddem N, 2019) *The Impact of an Augmented Reality Application on Learning Motivation of Students*. This study aims to determine the impact of using augmented reality mobile applications on the learning motivation of undergraduate health science students at the University of Cape Town. To explain motivation in the context of learning, the intrinsic motivation hypothesis was applied. The instructional materials motivation survey was utilized to develop the research instrument, and the attention, relevance, confidence, and satisfaction (ARCS) model guided the understanding of the impact of augmented reality on student motivation.

(Khan T, Johnston K, Ophoff J, 2019) *The integration of augmented reality in alternative learning application shows a promising effectiveness of integrating modern technology in education for 21st-century learners*. Such as the integration of learning nutritional concepts with augmented reality for kids aged 9-6 years old.

(Özerbaş, Demet, 2019) *The effects of marker-based augmented reality applications on academic achievement and permanence assessed the impact of learning objects developed in a marker-based augmented reality environment on academic achievement and the retention of learning*. An experimental design with pre-test and post-test control group was used to conduct the study.

2.5 Existing Nutrition Learning Applications

An existing application is available on the internet that provides nutritional facts about various foods with various categories. The content of this application provides basic reports of food. This includes the nutritional value that food can produce. (Alexey Korobov, 2015) This application gives numerous amounts of nutritional information that is overwhelming at a first look if the user does not know what they are looking into this application.

The nutrition learning application developed by the researchers is child friendly and can be easily comprehended by the users. Basic nutritional concepts are being provided on the developed application with interactive games and voice-over for the users. Another edge of the developed project is the integration of augmented reality which renders the application in an advantageous stage with promising results. As discussed, the use of modern technology such as augmented reality in education increases the motivation for learning. On the other hand, an existing application for nutrition quizzes has also been developed and is deployed on the google play store. This application has 160 questions and does not have instructions and lessons that can be used to answer the questions within the application. (Bayatov, 2018) The developed application for this study provides lessons within the application that helps the users answer the quiz questions within the application, the users are tasked to scan the 2d image of food in the application. Voice-over child-friendly descriptions are then heard from the application.

2.6 Current Conventional Teaching of Nutritional Facts

The Department of Education Region III in the Philippines has launched health and nutrition programs for public kindergarten and elementary learners in Central Luzon as part of its prevention measures against COVID-19. These programs aim to promote healthy habits among learners, even while they are at home, and include the school-based Feeding Program and the Distribution of Learners' Hygiene Kits.

The school-based Feeding Program targets public kindergarten students and Grades 1 to 6 students who were previously categorized as "wasted" or "severely wasted" based on last year's health records. The feeding program will provide nutritious food products and fresh or sterilized milk for at least 60 feeding days and has been implemented in public elementary schools since the last week of November. Dra. Gladys Lourdes Bengco, DepEd Region III Medical Officer IV, highlighted the importance of these programs in promoting good health and producing better learning for learners, even amidst the pandemic.

2.7 Summary

It is established that integrating mobile-based augmented reality learning resources for kids promotes involvement and is an excellent way to introduce augmented reality to the classroom. On the other hand, if a mobile application is attractive, kids will give it their full attention and it will be an effective learning tool for kids on the Android platform, resulting in a complete learning approach. Using the advantageous effects provided by modern technology in the field of education unlocks the potential of technology as an extension of alternative learning with amplified ways of imparting knowledge to learners.

The researchers of this project have seen the potential of modern technology such as augmented reality in the field of education, and the use of this as an extensive powerful tool in teaching and learning. The use of augmented reality in alternative learning applications demonstrates the potentially effective use of modern technology in education for learners of the twenty-first century.

2.8 Synthesis

In this section the researchers gathered existing applications that also have nutritional concepts which aim to inform the users. Existing applications and Researchers' applications are being compared with the following features. Augmented Reality feature, Lesson Module feature, Interactive Games, Nutritional Quiz feature, Rewards Based feature, Food Pyramid feature.

Table 2.1 Synthesis (Nutri-App)

Application	Augmented Reality Feature	Lesson Module Feature	Interactive Games	Nutritional Quiz Feature	Reward Based Feature	Food Pyramid
"Nutrition and Diet Quiz: Test Your Knowledge"				✓	✓	
"Nutrition Facts"		✓				✓
Nutri-App (Proposed Project)	✓	✓	✓	✓	✓	✓

CHAPTER 3

TECHNICAL BACKGROUND

The technicalities required to build the proposed application are discussed in this chapter. This includes the conceptual framework and software requirements.

3.1 Conceptual Framework

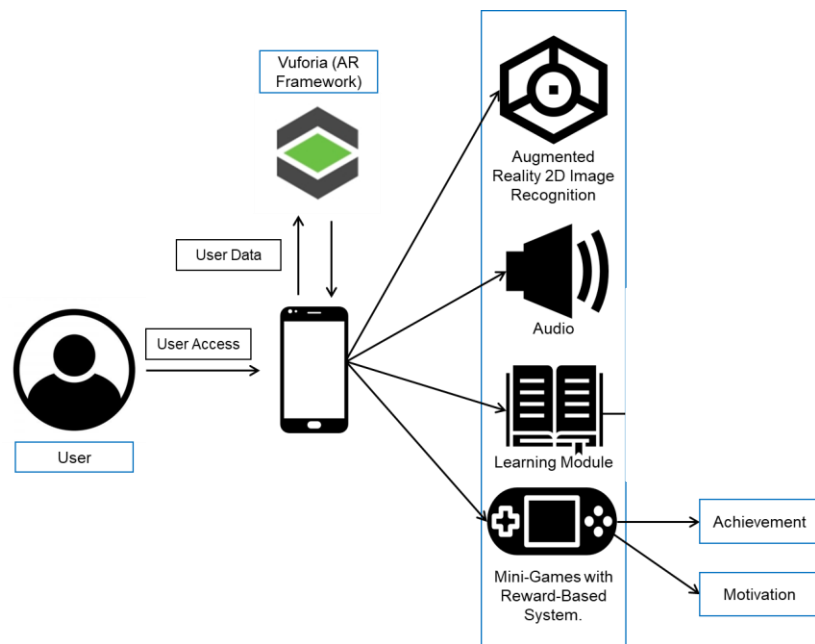


Figure 3.1 Conceptual Framework

In the figure above, The application will have multiple buttons for the user to use. In using the Image recognition feature the captured 2D image will be stored in the Vuforia Engine. Learning modules will be available for re-reading informational facts about nutritional foods. Mini-games will also be available for the users to play, having multiple games that will further enhance the curiosity and learning of the users.

3.2 Software Requirements

The following is the list of software requirements used for the development and implementation of the system.

3.2.a Programming IDE

- Unity - is a cross-platform game engine created by Unity Technologies that includes an IDE. It is employed to create video games on desktop platforms, web plugins, mobile phones, and consoles.

3.2.b Programming Language

- C# - (C-Sharp) is a programming language created by Microsoft that works with the .NET Framework and is used to create games, desktop applications, mobile applications, and web applications.

3.2.c Software Developer Kit

- Android SDK- a software development kit developed by Google for the Android platform.

3.2.d Augmented Reality

- Vuforia - A software development kit (SDK) Vuforia Engine is used to create apps for augmented reality. Advanced computer vision is easily incorporated by developers adding picture and object recognition functionality to any program and engaging with real-world settings.
- AR Foundation - allows users to work with augmented reality platforms in a multi-platform way within Unity. This package presents an interface for Unity developers to use but doesn't implement any AR features itself.

3.3 Hardware Requirements

The following is the list of hardware requirements needed by the user to be able to use the application. The hardware requirements for the development phase are also found in this section.

Hardware used by the developer for the development phase:

- Computer/Processor: Intel Core i3 & Above
- Memory/RAM: 8GB & Above
- Operating System: Windows 8 & Above

Hardware requirements for the user (mobile):

- Android Version 9 and Above

CHAPTER 4

DESIGN AND METHODOLOGY

In this chapter, the researchers gathered data on requirements analysis, requirement documentation, and design of the system software of the Nutri-App: Nutrition learning application using augmented reality for kids.

4. Requirement Analysis

To prepare for the development phase, the development team analyzed the collected data and evaluated which components to include in the documentation and the necessary modules to develop. The researchers used the learning material provided by DepEd as the basis for the application's content. They meticulously scrutinized the gathered data to select the essential elements that will be incorporated into the application. By carefully analyzing the collected data and selecting appropriate components, the development team was able to create a comprehensive and effective learning tool.

4.1 Architectural System Design

The figure below shows the conceptual model that defines the structure, behavior, and views of the Nutri-app.

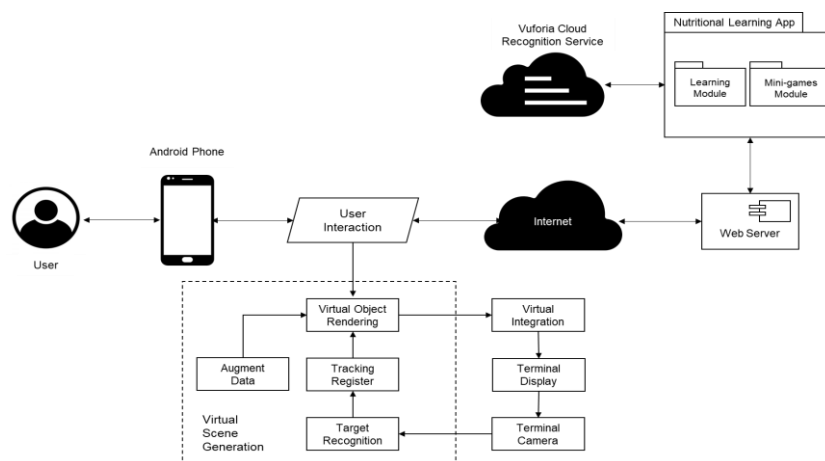


Figure 4.1 - Architectural System Design

In figure 2 shows the architecture system design of this project. The process of one of its main features is shown. An image target is uploaded on the Vuforia engine, to register it on the database. A plug-in of the vuforia engine is then integrated into the Unity 2d. For the application can recognize the target image, it must correspond to the uploaded image as the image target. With the vuforia software development kit, an advanced computer vision functionality application is integrated.

4.2 Flowchart

The diagram below presents the flow of the application from the user's view. It describes how the users can navigate through the application.

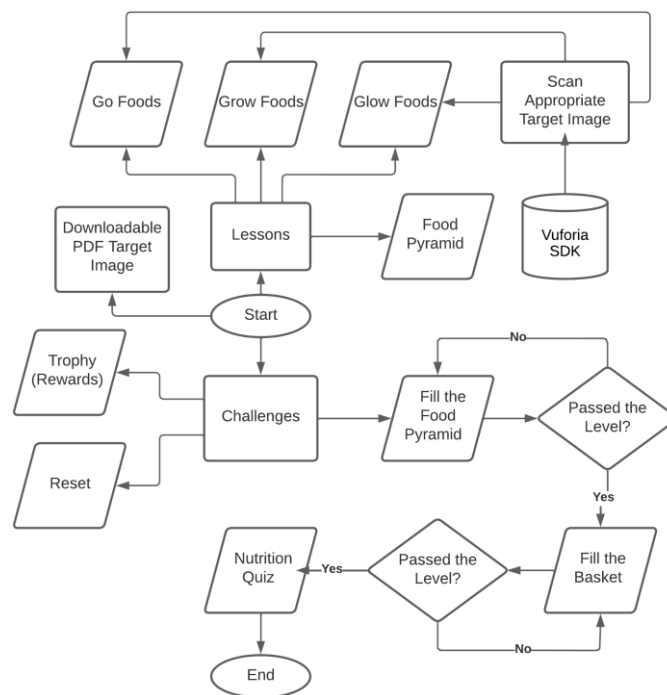


Figure 4.2 - Flow Chart

The figure above shows the flow of the system. How the users will be able to use the system with its constriction and rules. From the beginning of the system, the users will be able to see different buttons that will either lead them to image recognition, learning modules, mini-games, and scores.

The process of unlocking the mini-games requires the users to scan at least 5 images that can be found in the food pyramid. Resulting in an augmented reality image that shows brief nutritional facts of the scanned image and where part of the food pyramid is located.

4.3 Use Case Diagram of Nutri-App

The diagram below shows how the users will interact with the system. The user represents the player or learner class.

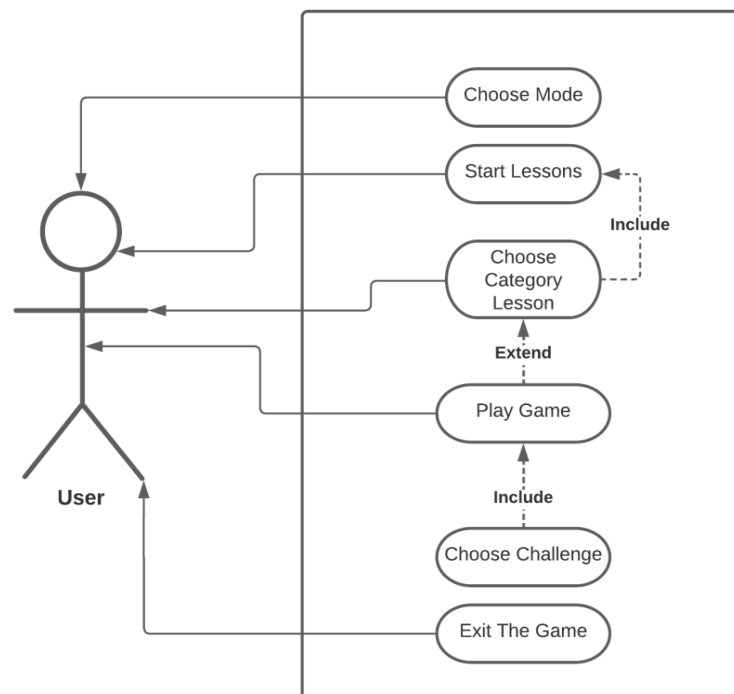


Figure 4.3 - Use Case Diagram of Nutri-App

In the figure shown above is the use case diagram. The users can choose two modes on the main menu. The Lessons and Challenges. The users can also exit from the game right away when they are on the main menu.

4.4 Requirements Documentation

The purpose of requirements documentation is to define what is needed for the application. The purpose of the application and its goals are discussed in this section.

4.4.1 Purpose

On this section presents the software requirements, features, and other related information about the Nutri-App: Nutrition Learning Application Using Augmented Reality for Kids. This application was developed for learners aged 6-9 years old.

This will help the learners to understand the concept of nutritious food in a fun way with the integration of marker-based augmented reality. This application can help increase user engagement with the help of fun and interactive games.

4.4.2 Goals

The main goal of this application is to provide an alternative way of learning nutritional concepts with the help of interactive games. Integrated with augmented reality to boost the interest of the learners or users.

4.4.3 Project Scope

This project is limited to the selected students aged 9-6 years old in Zamboanga City. The teachers can use this application as an alternative learning tool. The content and the lessons of the application are based on the learning materials provided by the Department of Education. The lesson mode of the application has been integrated with augmented reality with a voice-over about the particular target image.

On this feature, Glow food, Grow food, and Go food are found. The first challenge from the challenge mode is fill-out the food pyramid, followed by the fill-in-the-basket, and lastly the quiz game.

The main purpose of the challenges is to assess the learned content of the learners from the application rewarding them with a virtual trophy.

4.4.4 Success Criteria

The Nutri-App: Development of a nutrition learning application for kids using augmented reality. This application will be considered a success if:

- The main page of the application is working and without error.
- The application must meet all the requirements.
- The Augmented Reality API is working without error.

4.5 User Characteristics

Categories of users can be distinguished for this application:

- Learners (aged 6-9 years old) - The main users of this application, needs to learn the nutritional concepts in a fun and interactive way.
- Grade school Teacher - This user can use the application for the purposes of teaching or learning.

4.6 Mandated Constraints

- The capstone team of this project used c# for the development phase.

4.7 Non-Functional Requirements

4.7.1 Usability Requirements

- Users can access the downloadable PDF file of the target images on the main screen.

4.7.2 Operational Requirements

- All users can access the lesson and the challenges subsequently.

4.7.3 Performance Requirements

- On start-up, the application should load for at least 5 seconds.

4.7.4 Other Quality Attributes

- The usability of the application was designed as child user-friendly for ease-of-use of the application, without the need for assistance from experts.

4.7.5 Documentation and Training

- Training will not be conducted.

4.8 Functional Requirements

FR1: The system should be able to track images for augmented reality in lesson mode.

FR2: The system should be able to save the screenshots

FR3: The system should be able to have a voice-over when an image target is being hovered.

FR4: The system must be able to respond to simple user navigation.

4.9 System Features

4.9.1 Main Menu - On the main page, the users should be able to choose from the two main buttons. The Lessons and Challenges button. On the lower part of the main screen, a button for the downloadable PDF of the image targets is accessible.

4.9.2 Lessons - On this main feature, the learners have access to the 3 buttons for the Glow food, Go food, and Glow food. That has the embedded API of the vuforia augmented reality.

The users can scan the corresponding target image in accordance with the clicked image on each category. The last feature for this section is the food pyramid, the users can click this and have the actual food pyramid on the screen.

4.9.3 Challenges - On the second main feature of this application is the Challenges section. Wherein the users can access the games subsequently. The following challenge will remain locked if the user is not able to complete the challenge. The first challenge is the Fill the Food Pyramid Challenge, wherein the users must drag and drop the food image onto the silhouette of its shape. The second challenge is the Fill-in-the-Basket, the users must fill in each basket with the correct food that belongs to the category of Glow, Go, and Grow. The last challenge is the Nutrition Quiz, on this challenge the users must answer the questions correctly for them to receive a reward.

4.9.4 Progress Reset - This button can be found in the Challenges section. The users are allowed to reset the progress of their challenges.

4.10 Design of Software System

In order to provide orientation to the development team regarding the system architecture, the development team will keep track of the data needed for the defined architecture and system design efficiently.

4.10.1 Process

The navigation to use the application is child-friendly as the main users are learners aged from 6-9 years old. The overall design for the application was made in a way that the users can navigate through the application with ease. We used unity 2d for the development with Vuforia API to integrate the augmented reality feature of the application. For the application's user interface, we used Adobe XD, and the images on the application are from open-source Canva Education.

4.10.2 Data Flow Diagram (DFD)

a. Level 0 - Context Diagram

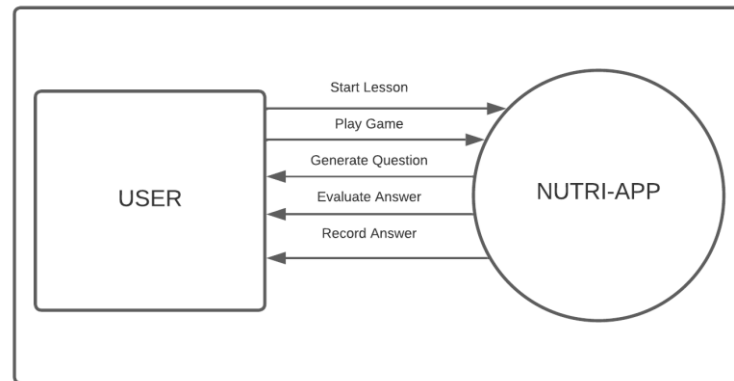


Figure 4.3 - Context Diagram (Level 0)

The figure above shows the at-first-view of the application as a single-high-level process that offers the overview of the external entities towards the application.

a. Level 1 - Data Flow Diagram

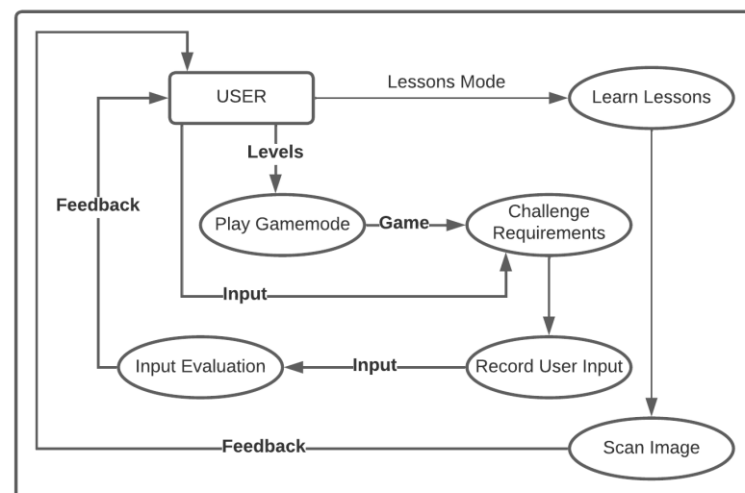


Figure 4.4 - Data Flow Diagram (Level 1)

In level 1 DFD, the single process node from the context diagram is broken down into sub-processes. The “exploded view” of the context diagram.

CHAPTER 5

DEVELOPMENT AND TESTING

This chapter describes the graphical user interfaces, implementation plan, and the different testing of the developed application Nutri-App: Nutrition Learning Application using Augmented Reality for Kids.

5.1 Description of Prototype

In creating the application prototype, the researchers of this project made a structured process for the development team as a guide. The conceptualization of the application user interface was established first. The color palette, images, game concepts, and font styles. As these are the essential components of the overall design of the application. A wireframe for the prototype was created on adobe XD for the convenient creation of the actual user interface in the later part of the development. The choice of font, image design, and colors for this application was of crucial importance for the impact of this application on its main users, aged 6-9 years old. The font for this application was child-friendly and appealing to the sight of every user. Image designs were of the same art style to create consistency for the overall design of the nutritional concept.

5.2 Integration of Augmented Reality

The programming language used by the developer of this project is C#. Unity for the programming IDE is used to create the two-dimensional games on the application. The developer used Vuforia for the software development kit to identify an image that triggers the appearance of 2d objects created in unity using marker-based augmented reality.

For integrating Augmented Reality in lesson mode, the developer follows a step-by-step process:

1. A sample model for AR triggers and objects was created
2. Add the license key to a Unity Vuforia config.
3. Create a target manager and a database.
4. It generates a database by importing AR triggers.
5. Choose Unity as the platform, download the database, and import AR triggers into it.
6. Import downloaded packages into Unity.

5.3 Graphical User Interface (GUI)

The Nutri-App is a Nutrition Learning Application that has been specifically designed for kids. The user interface of this application comprises of figures that are visually attractive and have a child-friendly appearance. These figures have been carefully selected to engage children and to create an interactive learning experience. The use of augmented reality in this application further enhances the learning process, making it more immersive and enjoyable for children. Overall, the user interface of the Nutri-App has been crafted to appeal to children, and to create an environment that encourages them to learn about nutrition and healthy eating habits.



Figure 5.1 - GUI Main Screen and Lessons

Figure 5.1 serves as the primary graphical user interface of the project, showcasing the main screen of the application. This screen displays two primary buttons, namely the "Lessons" and "Challenges" buttons. These buttons are prominently displayed and serve as the main entry points for users to navigate through the application. By clicking on the "Lessons" button, users are taken to a secondary graphical user interface, which is displayed in Figure 5.2. This interface provides access to the various lessons available in the application. require.

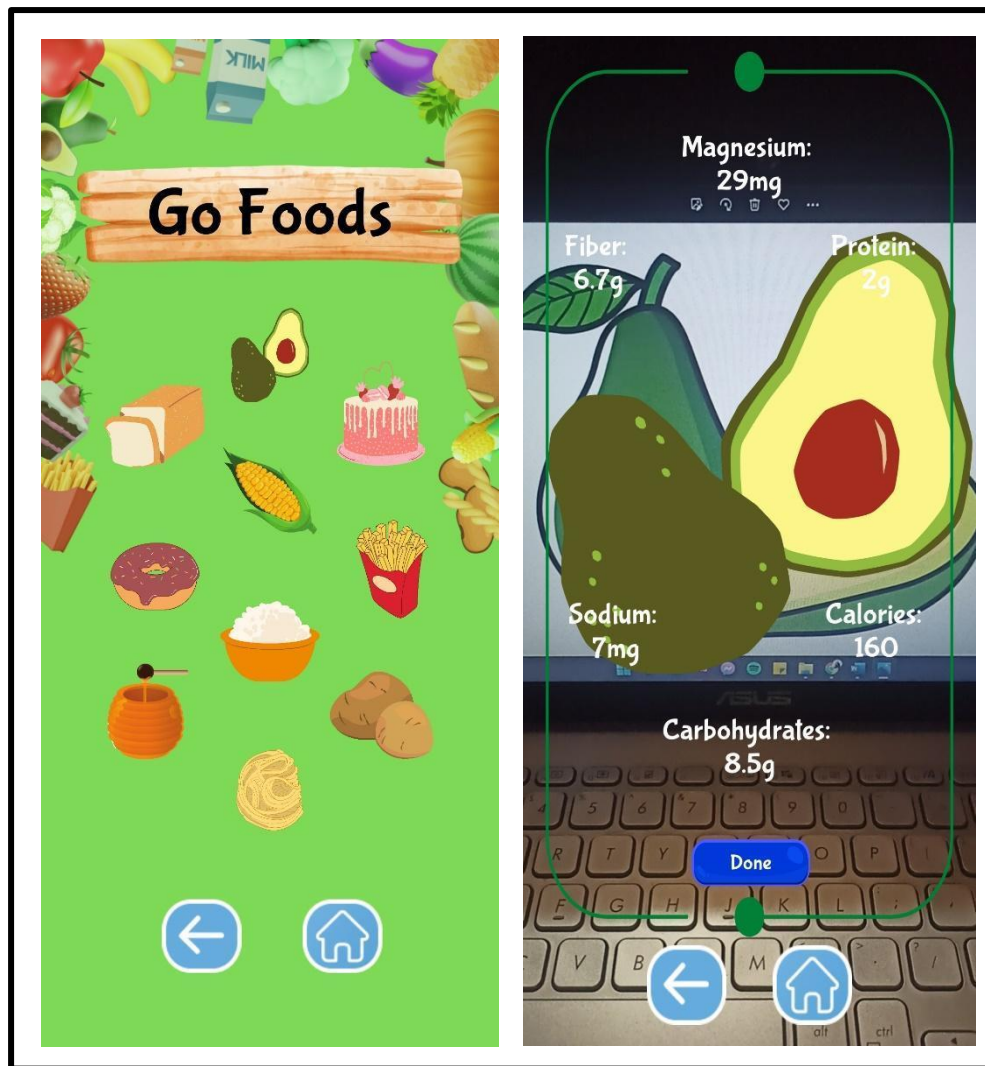


Figure 5.2 - GUI Go Foods and Augmented Image

Figure 5.3 displays the graphical user interface that appears when the user selects the "Go Foods" button in the application. The interface presents ten different food options that the user can choose from and scan. This feature allows users to learn about different foods and their nutritional value in an interactive way. Upon selecting and scanning a food item, the user is directed to Figure 5.4, which showcases the scanned image using marker-based augmented reality technology. This feature provides an immersive experience for users, allowing them to see a virtual representation of the food item they scanned.

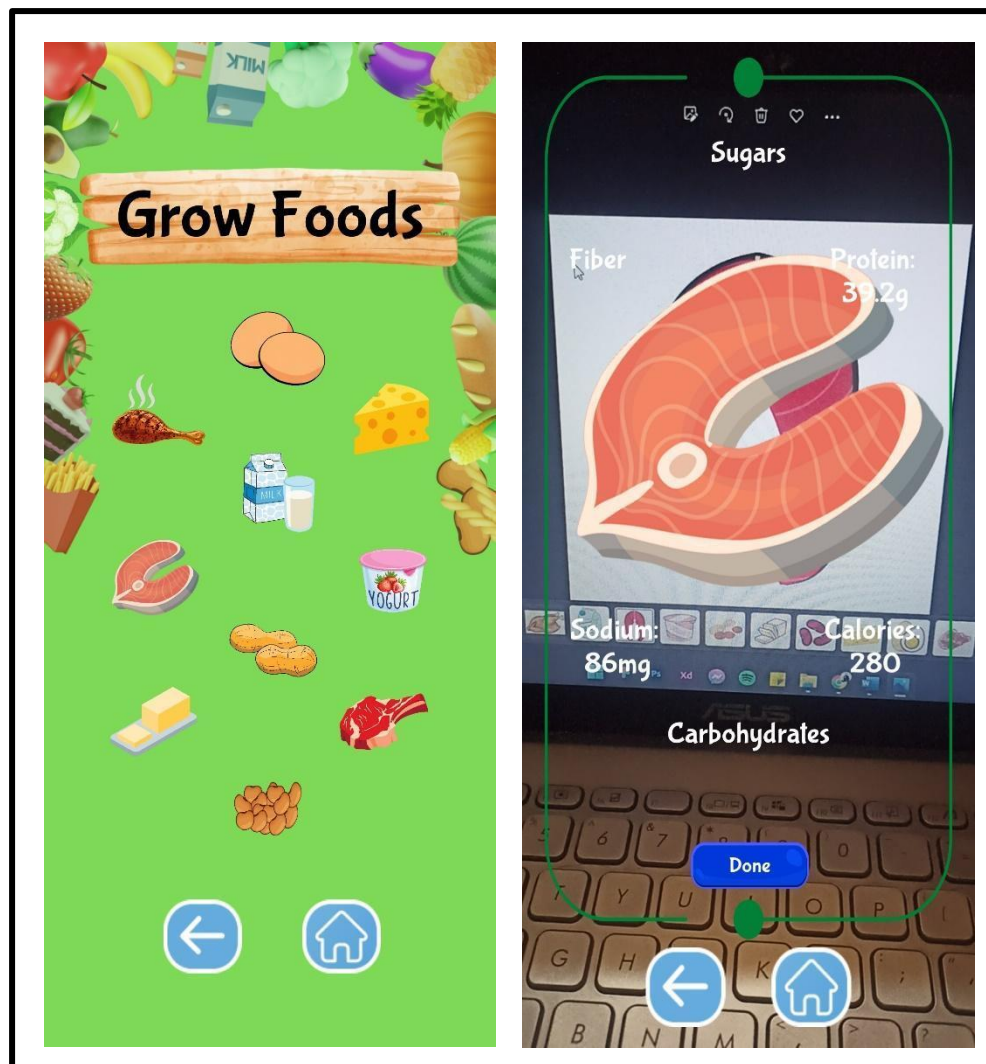


Figure 5.3 - GUI Grow Foods and Augmented Image

Figure 5.5 is a graphical user interface that is displayed when the user selects the "Grow Foods" button in the application. This interface presents ten different food options that the user can choose from and scan.

The selection of these food items is based on their nutritional value and their ability to promote growth and development in children. Once the user selects a food item, they can scan it using their device's camera, which leads them to Figure 5.6. This graphical user interface showcases the scanned image of the food item using marker-based augmented reality technology.

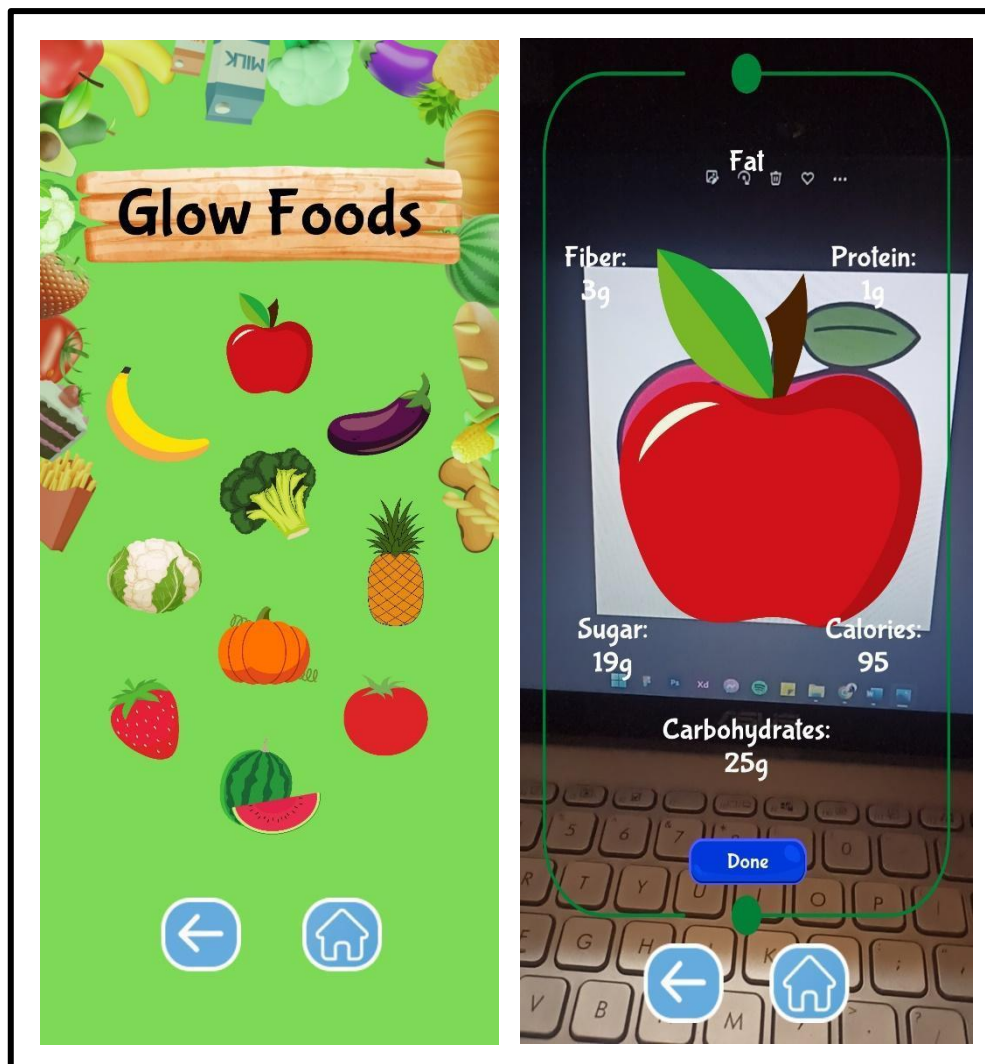


Figure 5.4 - GUI Glow Foods and Augmented Image

Figure 5.7 is a graphical user interface displayed when the user selects the "Glow Foods" button in the Nutri-App. This interface presents ten different food options that the user can choose from and scan. These food items were selected based on their ability to promote healthy skin and eyesight in children. Once the user selects a food item, they can scan it using their device's camera, which leads them to Figure 5.8. This graphical user interface shows the scanned image of the food item using marker-based augmented reality technology. This feature allows users to view a virtual representation of the food item, while also learning about its nutritional benefits.

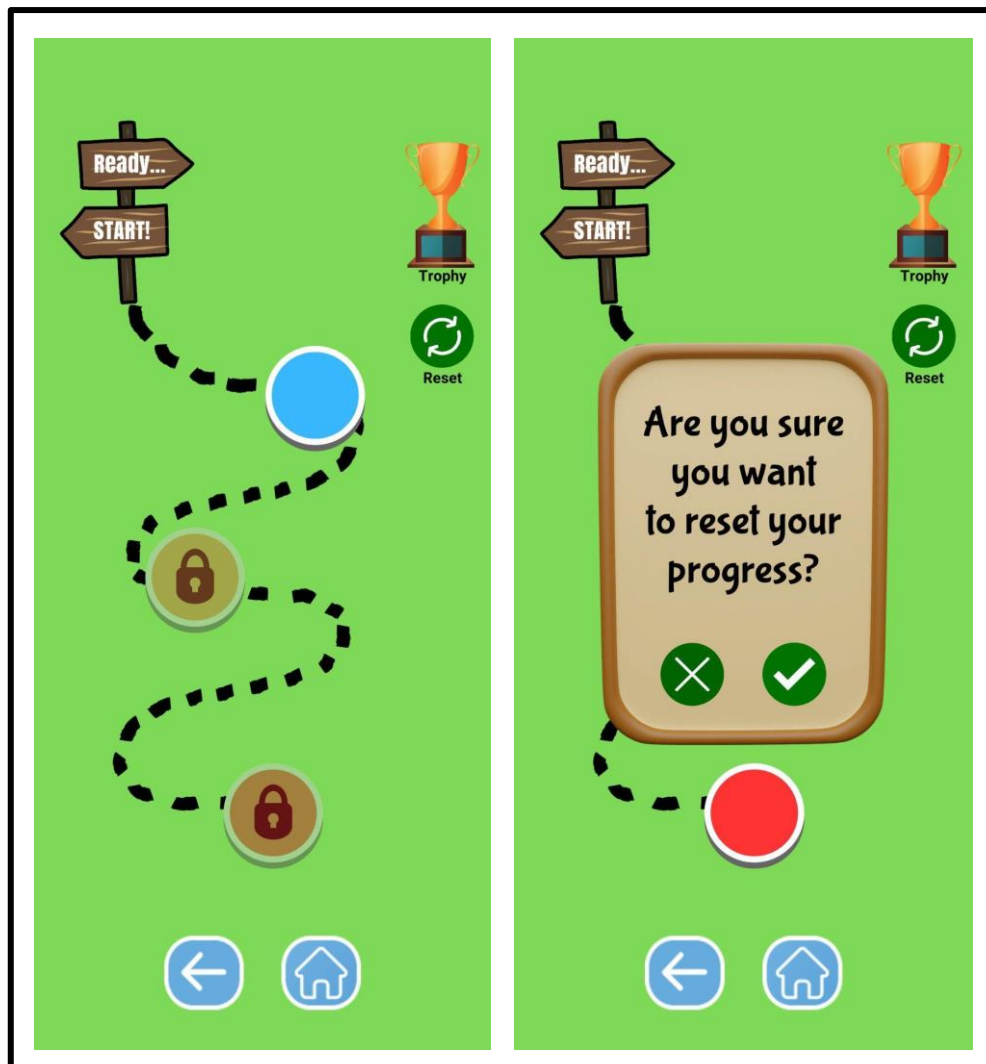


Figure 5.5 – Challenges and Quiz Reset

Figure 5.9 displays the graphical user interface of the challenges/games for users in the Nutri-App. This interface presents a series of interactive games that users can play to reinforce their knowledge of nutrition and healthy eating habits. As the user progresses through the games, more challenges will be unlocked, providing an engaging and challenging experience that promotes learning. Figure 5.10 shows the reset feature of the challenges, which allows users to reset their progress and start again from the beginning.

This feature is useful for users who want to revisit previous challenges or for new users who want to start from the beginning. By offering a reset feature, users can revisit past challenges and reinforce their knowledge of healthy eating habits, leading to better retention of the information.

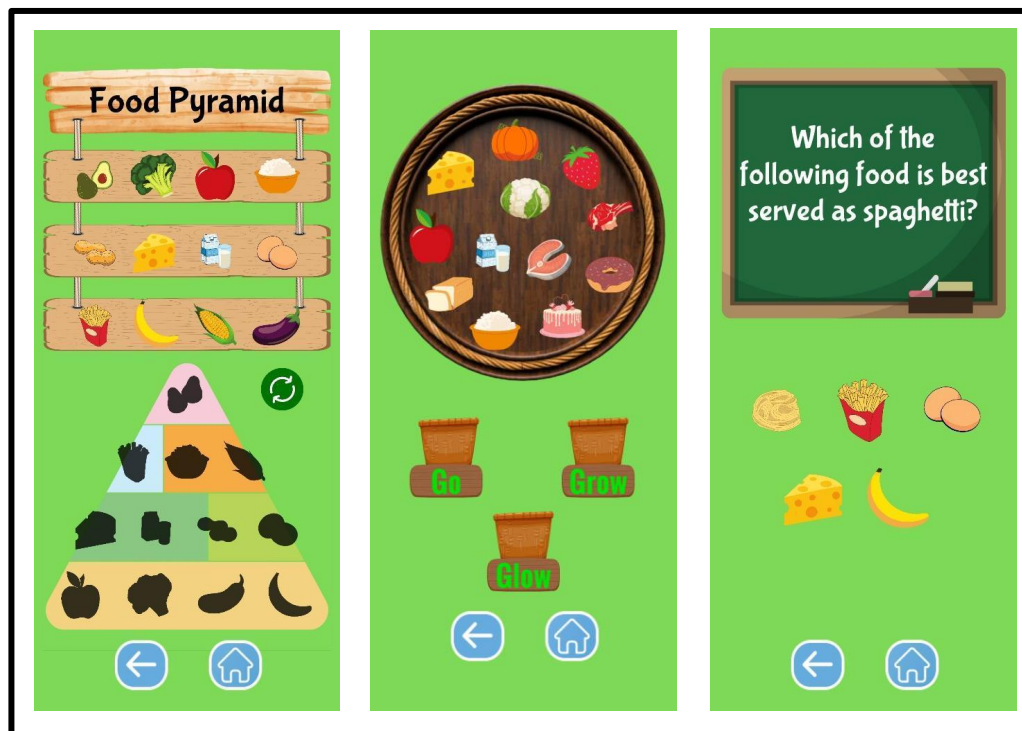


Figure 5.6 - Nutri-app Games

In Figure 5.11, we can see the graphical user interface of the first game called Fill-in the Food Pyramid, where users have to drag and drop different food items into their proper places on the food pyramid. Figure 5.12 shows the second game, which is called Fill the Basket, where learners must drag and drop items into the appropriate basket based on their food group. In Figure 5.13, we can see a sample question from the game challenges, which the users have to answer to progress in the game.

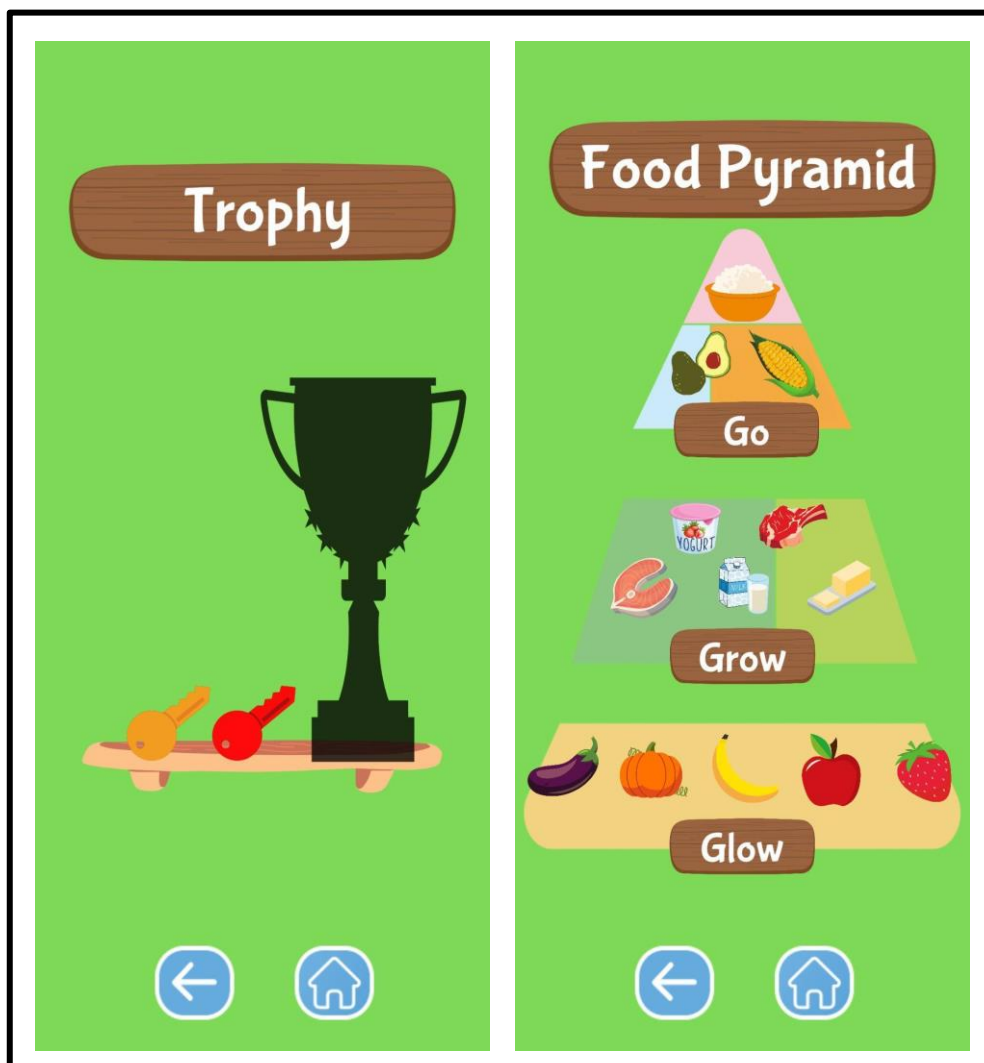


Figure 5.7 – Collection of Tokens and Food Pyramid Lesson

Figure 5.14 displays the graphical user interface of the keys and trophy icons which are obtained by users when they successfully complete a challenge. The keys represent the number of challenges completed while the trophy indicates a certain level of achievement. On the other hand, Figure 5.15 showcases the food pyramid, which is part of the lesson feature of the Nutri-App. It provides users with an interactive way of learning about the different food groups and their corresponding nutritional value.

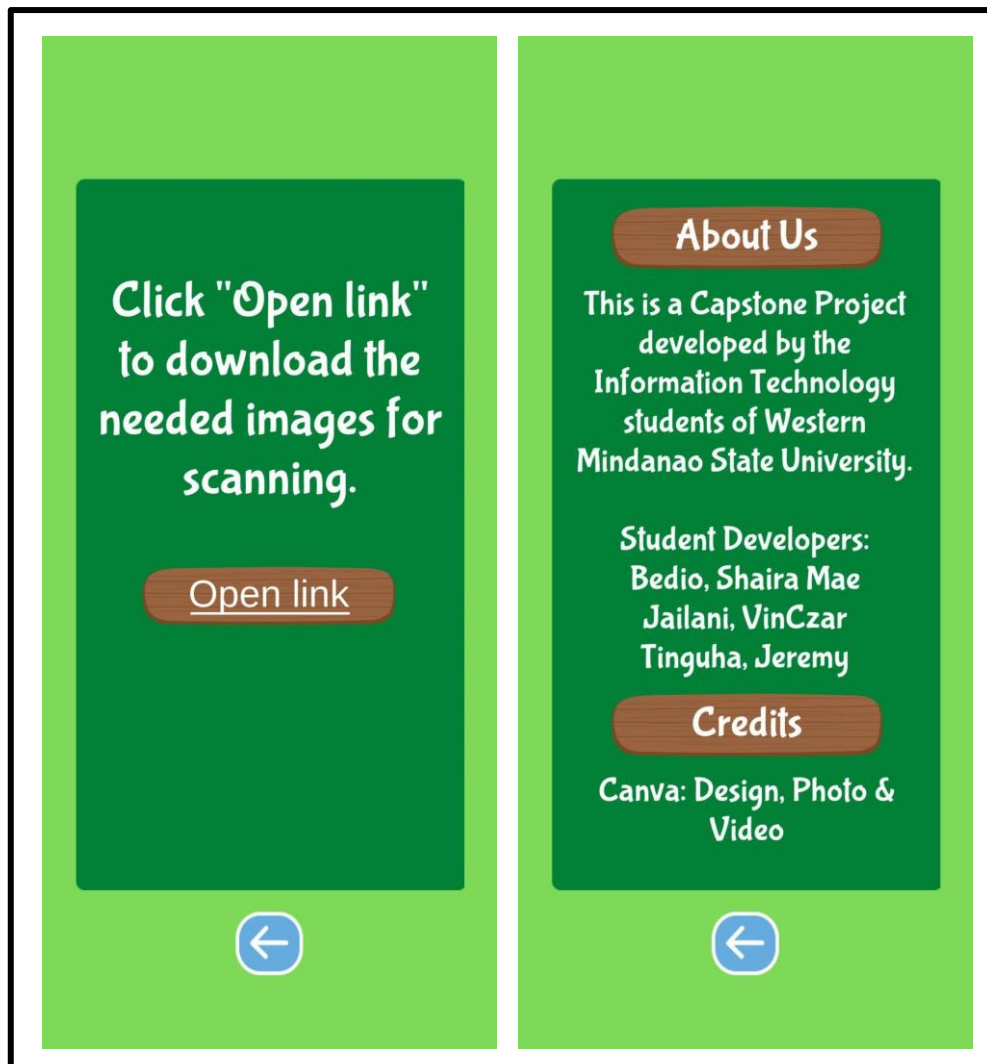


Figure 5.8 – Downloadable Link and About Us

Figure 5.16 displays the user interface for the downloadable PDF file that contains the target images used in the marker-based augmented reality feature of the Nutri-App. These target images are essential for the app's functionality and can be printed out by the user to enhance their experience. Figure 5.17 provides brief information about the capstone team responsible for developing this project, giving credit to the individuals who worked on the app's design and development.

5.4 Development

The Developers used the Scrum Agile Development Method. Scrum is an Agile development system that employs sprints, which are fixed-length work iterations. This method uses a scrum board to represent all of the work completed during a sprint; this board can display many steps in the process, including To Do, In Progress, and Done. The Scrum method is composed of five stages. Initiate, Plan and Estimate, Implement, Review and Retrospect and lastly Release.

- Initiate – This phased process is the initiation of the project. Identifying the scrum master, and scrum team, creating of the priority product backlog, and conducting the release planning.
- Plan and Estimate – This phase consists of a process related to planning and estimating tasks. Includes creating tasks, estimating tasks, and creating sprint backlog.
- Implement – This phase executes the task and activities to create a product. This phase includes sprint planning, sprint demo, daily stand-up, and retrospective which is conducted every day or week or sprint.
 - Sprint Planning – Team planning meeting that determines what to complete in the coming sprint.
 - Sprint Demo – A meeting where the team shows what they have shipped in the sprint.
 - Daily Standup – A mini-meeting for the software team to synchronize work.
 - Retrospective – A review of what did and did not go well with actions to make the next sprint better.
- Review and retrospect – This phase review the deliverables and the work that has been done.
- Release – This phase delivers the accepted deliverable to the customer and identifies and documents the lesson learned during the project.

5.5 Testing

To demonstrate the application's usability, the researchers put a number of tests. The test's objective is to evaluate the application's effectiveness. To determine whether the developer complied with all the criteria necessary to assess the application's success.

5.5.1 Unit Testing

This test aids in the developer's comprehension of the code. In order to make changes more quickly in the event of a defect or error, it enables developers to evaluate the validity of the developed code and test the correct functionality of each module. The application has 60 unity scenes with code scripts. The two main modes are the lesson mode and the challenges mode. In the lesson mode, there are 3 modes for the scanning of marker-based augmented reality. The challenges mode is comprised of challenges, starting from Fill-out the Food Pyramid, Fill the basket, and Nutrition Quiz.

Table 5.1 Unit Testing

ID	Test Case	Expected Result	Pass/ Failed
1	User starts the application.	The user is on the loading screen and then the main menu.	PASS
2	User selects the lesson mode.	The user is in lesson mode with 4 options.	PASS
3	User selects the glow foods category.	The user is in the glow foods category, with 10 various food to scan.	PASS
4	User selects the go foods category.	The user is in the go foods category, with 10 various food to scan.	PASS
5	User selects the grow foods category.	The user is in the grow foods category, with 10 various food to scan.	PASS

6	User selects the food pyramid.	The user is the lesson mode of the food pyramid.	PASS
7	User selects the challenges mode.	The user is in subsequent games once passed/unlocked.	PASS
8	User selects the first challenge. (Fill-out the Food Pyramid)	The user is in the fill-out-the-food pyramid challenge.	PASS
9	User selects the second challenge. (Fill-the-Basket)	The user is in the fill-the-basket challenge.	PASS

5.5.2 User Interface Testing

The purpose of this test is to let the researchers confirm that the application's functionalities are operational. 10 learners and 10 teachers/student-teachers were gathered by the researchers of this project to test the functionality of the application. Functions like buttons, colors, and fonts, as well as any visual cues and graphical icons, and menus. In Learners evaluation of the nutri-app in terms of overall aspect, all of the thirty respondents were able to respond to the evaluation and came to a conclusion that they like learning from the application.

5.5.3 User Acceptance Testing

An end-user evaluation was conducted by the researchers on the respondents where they were all able to use the working application and test the functionality that is proportional to the objectives of the application. The teachers and student-teachers reached a consensus that incorporating augmented reality into the application had a positive impact on the students' desire to learn. The data gathered from the study clearly demonstrates that using augmented reality improved the students' motivation when it came to grasping nutritional concepts through the application. Additionally, the teachers' feedback highlighted the significance of this application as a crucial topic within the curriculum.

5.6 Implementation Plan

The project was developed in 150 days from May 2022 to November 2022. Started the development at the designing of the user interface to the end of the development phase. The iterative pattern for early-stage processes was established from the beginning of the development phase. First, the researchers proposed a project to the subject adviser. When the proposed project was approved, the researchers started to conduct the study. There were two types of questionnaires for the two different types of respondents. The teachers and the learners.

Three categorized plans were devices for the implementation phase. First is data collection from the respondents through google forms. Second is the user interface design, the development team conducted a brainstorming session on what design, font, and art style are suitable for the overall concept of the application.

Lastly, is the development phase, collecting all of the required information from the beginning of the implementation phase, and later on is implemented in the development phase. Following a successful event, each module is tested several times to ensure that there are no errors. Unit testing is the initial evaluation, and all of the source code has undergone testing to ensure that the developed code is accurate from the unity scenes. The second test is integration testing, which combines all of the application's modules for testing. The final test is acceptability testing, in which the program was examined and reviewed by actual users.

CHAPTER 6

RESULTS AND DISCUSSIONS

This chapter represents the results and discussion of the Nutri-App: Nutrition Learning Application using Augmented Reality for Kids.

6.1 Introduction

The researchers of this project conducted data gathering through a questionnaire to have a thorough assessment of the developed application. After the development phase, the researchers conducted an assessment of the target users using end-user evaluation and acceptance testing. The collected data are from 40 specific users from Zamboanga City. There are 30 respondents for learners aged 6-9 years old, and 10 respondents for teachers. The evaluation questionnaires are composed of categorized questions. For the teachers' evaluation, the questions are divided into Instructional Content, Instructional Design, Management and Motivation, Ease-of-Use, and Technical Aspects. For the learners' evaluation, the questions are divided into Content, Designs and Sound, Engagement and Interactivity, and Overall assessment of the questionnaire.

Table 6.1 Number of Respondents (Learners)

Age	Number of Respondents
6 years old	10
7 years old	5
8 years old	7
9 years old	3
Total Respondents	30

The table above shows the number of respondents in accordance with their age. The collected data was taken from 30 respondents for the learners' evaluation.

Table 6.2 Number of Respondents (Teachers)

	Number of Respondents
Student-Teacher	6
Teacher	4
Total Respondents	10

The table above shows the number of respondents in accordance with their teaching status. The collected data was taken from 10 respondents for the teachers' evaluation.

6.2 Results of the Implementation of Nutri-App

The evaluation was facilitated using printed evaluation forms and a demo of the Nutri-app. To gather data for user-acceptance testing, a face-to-face evaluation was conducted with learners, supervised by either teachers or parents. The data collected from the evaluation were presented in the figures below, which provide an interpretation of the respondents' feedback on each objective. The study aimed to achieve four distinct objectives, which are reflected in the data collected during the evaluation process.

6.3 End-User Application Evaluation towards Nutri-App (Learners)

This end-user evaluation is done after the end-users tested the application Nutri-app: Nutrition Learning Application Using Augmented Reality for Kids. Based on the results of the evaluation, the respondents' view of the design, sound, and ease of use of the application is an engaging and interesting alternative learning tool. The lessons included in the application are beneficial to the nutritional knowledge of the end-users of the application.

6.3.1 Evaluation of Nutri-App in terms of Content

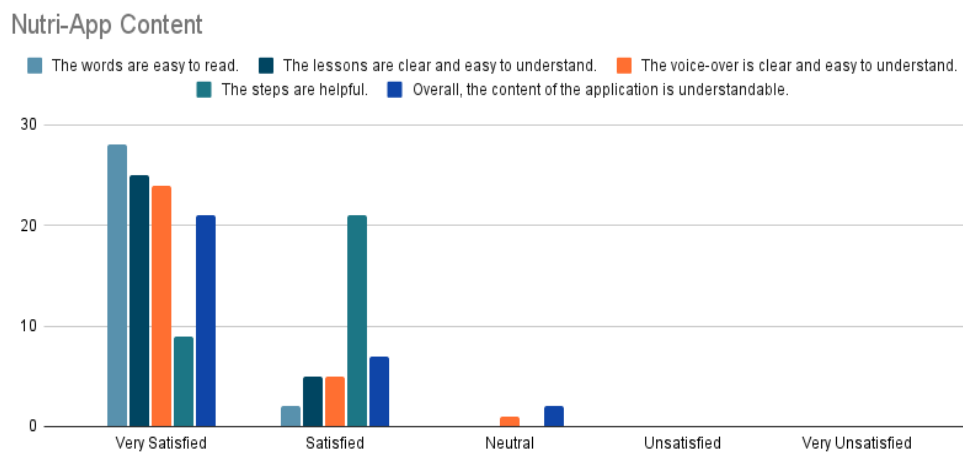


Figure 6.1 - Content Evaluation

The figure above shows the result of end-user evaluation in terms of the content of the application. Based on the feedback from the Learners:

In terms of the Nutri-app content, "The words are easy to read." 28 out of 30 respondents or 93.33% answered very satisfied. 2 out of 30 respondents or 6.67% answered satisfied. Moreover, in terms of the Nutri-app content, "The lessons are clear and easy to understand." 26 out of 30 respondents or 86.66% answered very satisfied. 4 out of 30 respondents or 13.34% answered satisfied. In terms of the Nutri-app content, "The voice-over is clear and easy to understand." 25 out of 30 respondents or 83.33% answered very satisfied. 4 out of 30 respondents or 13.34% answered satisfied. and 1 out of 30 respondents or 3.33% answered neutral.

In terms of the Nutri-app content, "The steps are helpful." 9 out of 30 respondents or 30% answered very satisfied. and 21 out of 30 respondents or 70.00% answered satisfied. Lastly, In terms of the Nutri-app content "Overall, the content of the application is understandable." 21 out of 30 respondents or 70.00% answered very satisfied. 7 out of 30 respondents or 23.33% answered satisfied. and 2 out of 30 respondents or 6.67% answered neutral.

6.3.2 Evaluation of Nutri-App in terms of Design and Sounds

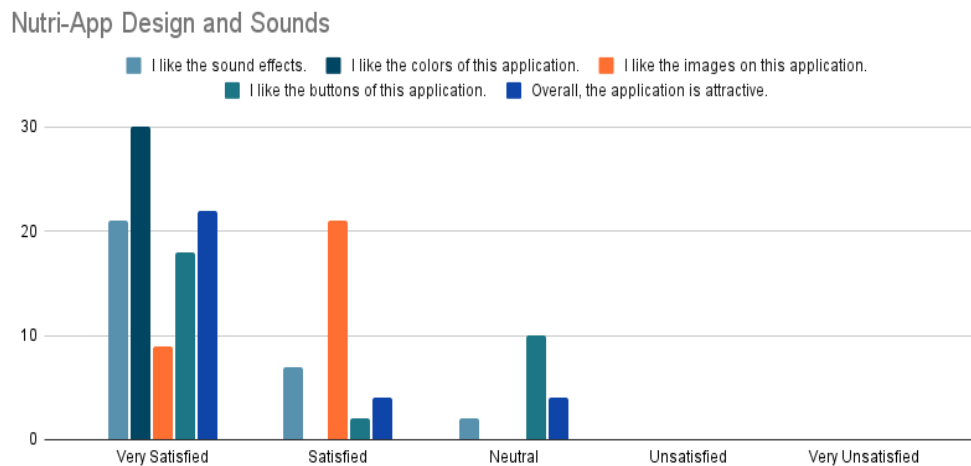


Figure 6.2 - Design and Sounds Evaluation

The figure above shows the result of end-user evaluation in terms of the design and sounds of the application. Based on the feedback from the Learners:

In terms of the Nutri-app design and sounds evaluation, "I like the sound effects." 21 out of 30 respondents or 70.00% answered very satisfied. 7 out of 30 respondents or 23.33% answered satisfied. and 2 out of 30 respondents or 6.67% answered neutral. Moreover, in terms of the Nutri-app design and sounds evaluation, "I like the colors of the application." 30 out of 30 respondents or 100.00% answered very satisfied.

Furthermore, in terms the Nutri-app design and sounds evaluation, "I like the images on this application." 9 out of 30 respondents or 30.00% answered very satisfied. 21 out of 30 respondents or 70.00% answered satisfied. In terms of the Nutri-app design and sounds evaluation, "I like the buttons of this application." 18 out of 30 respondents or 60.00% answered very satisfied. 2 out of 30 respondents or 6.67% answered satisfied.

And 10 out of 30 respondents or 33.33% answered neutral. Lastly, in terms of the Nutri-app design and sounds evaluation, "Overall, the application is attractive."

22 out of 30 respondents or 73.33% answered very satisfied. 4 out of 30 respondents or 13.33% answered satisfied. and 4 out of 30 respondents or 13.33% answered neutral.

6.3.3 Evaluation of Nutri-App in terms of Engagement and Interactivity

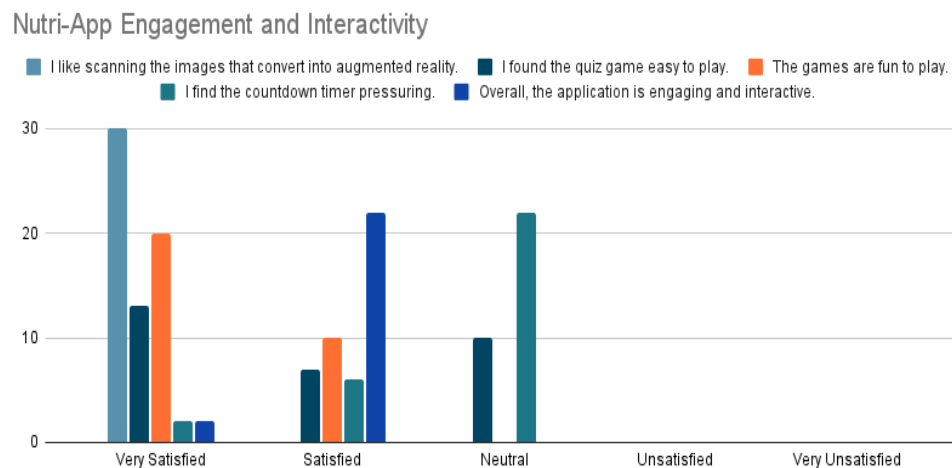


Figure 6.3 - Engagement and Interactivity Evaluation

The figure above shows the result of end-user evaluation in terms of the interactivity of the application. Based on the feedback from the Learners:

In terms of the Nutri-app engagement and interactivity evaluation, "I like scanning the images that convert into augmented reality." 30 out of 30 respondents or 100.00% answered very satisfied. In terms of the Nutri-app engagement and interactivity evaluation, "I found the quiz game easy to play." 13 out of 30 respondents or 43.33% answered very satisfied. 7 out of 30 respondents or 23.33% answered satisfied. and 10 out of 30 respondents or 33.33% answered neutral.

Furthermore, in terms of the Nutri-app engagement and interactivity evaluation, "The games are fun to play." 20 out of 30 respondents or 66.66% answered very satisfied. 10 out of 30 respondents or 33.33% answered satisfied. Moreover, in terms of the Nutri-app engagement and interactivity evaluation, "I find the countdown timer pressuring."

2 out of 30 respondents or 6.67% answered very satisfied. 6 out of 30 respondents or 20.00% answered satisfied. and 22 out of 30 respondents or 73.33% answered neutral. Lastly, in terms of the Nutri-app engagement and interactivity evaluation, "Overall, the application is engaging and interactive." 2 out of 30 respondents or 6.67% answered very satisfied. 28 out of 30 respondents or 93.33% answered satisfied.

6.3.4 Evaluation of Nutri-App in terms of Overall Aspect

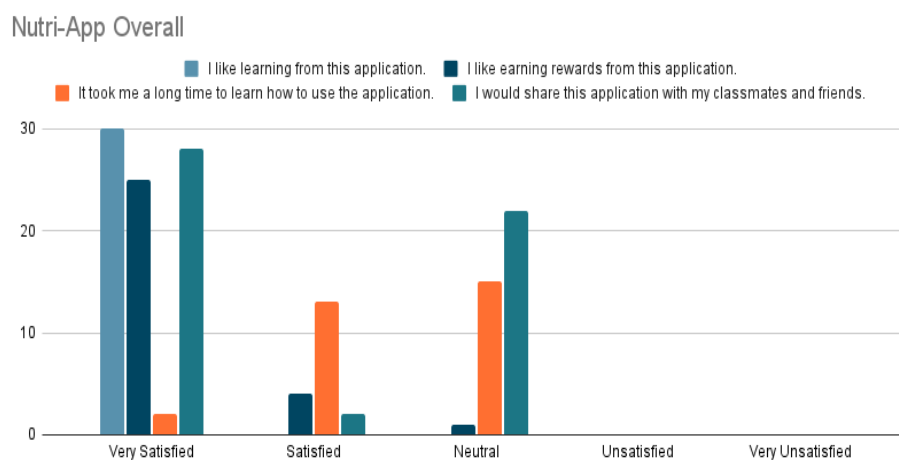


Figure 6.4 - Overall Evaluation

The figure above shows the result of end-user evaluation in terms of the overall content of the application. Based on the feedback from the Learners:

In terms of the Nutri-app overall evaluation, "I like learning from this application." 30 out of 30 respondents or 100.00% answered very satisfied. Moreover, in terms of the Nutri-app overall evaluation, "I like earning rewards from this application." 25 out of 30 respondents or 83.33% answered very satisfied. 4 out of 30 respondents or 13.33% answered satisfied. and 1 out of 30 respondents or 3.33% answered neutral.

In terms of the Nutri-app overall evaluation, "It took me a long time to learn how to use the application." 2 out of 30 respondents or 6.67% answered very satisfied. 13 out of 30 respondents or 43.33% answered satisfied. and 15 out of 30 respondents or 50.00% answered neutral.

Lastly, in terms of the Nutri-app overall evaluation, "I would share this application with my classmates and friends." 28 out of 30 respondents or 93.33% answered very satisfied. 2 out of 30 respondents or 6.66% answered satisfied.

6.4 End-User Application Evaluation towards Nutri-App (Teachers)

Teachers underwent a face-to-face evaluation as part of the user-acceptance testing. The information was presented in the figures below to interpret the respondents' answers to each objective.

6.4.1 Evaluation of Nutri-App in terms of Instructional Content

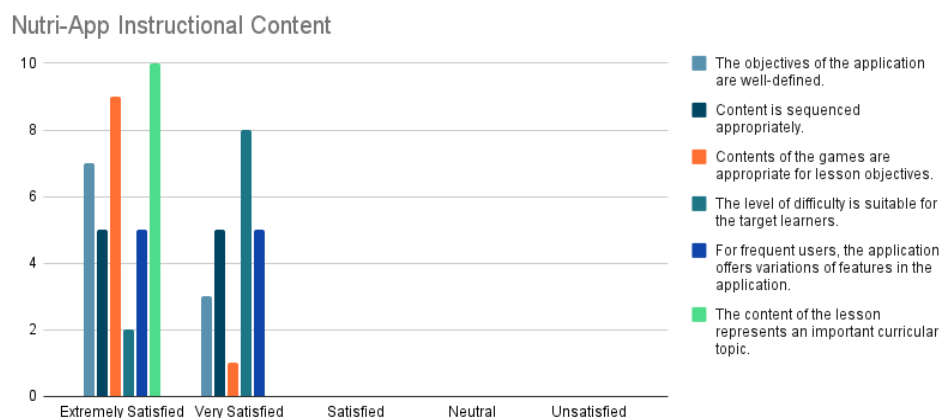


Figure 6.5 - Instructional Content Evaluation

In the figure above shows the Instructional Design of the application, assessed by teachers and student-teachers. Most of the respondents agreed that the contents of the games are appropriate for the objectives of the lesson. The objectives of the application are also well-defined.

In terms of the Nutri-app Instructional Content Evaluation, "The objectives of the application are well-defined." 7 out of 10 respondents or 70.00% answered extremely satisfied. 3 out of 10 respondents or 30.00% answered very satisfied.

In addition of the Nutri-app Instructional Content Evaluation, "Content is sequenced appropriately." 5 out of 10 respondents or 50.00% answered extremely satisfied. 5 out of 10 respondents or 50.00% answered very satisfied.

In terms of the Nutri-app Instructional Content Evaluation, "Content of the games are appropriate for lesson objectives." 9 out of 10 respondents or 90.00% answered extremely satisfied. 1 out of 10 respondents or 10.00% answered very satisfied.

Furthermore, in terms of the Nutri-app Instructional Content Evaluation, "The level of difficulty is suitable for target learners." 2 out of 10 respondents or 20.00% answered extremely satisfied. 8 out of 10 respondents or 80.00% answered very satisfied. In terms of the Nutri-app Instructional Content Evaluation, "For frequent users, the application offers various of features in the application." 5 out of 10 respondents or 50.00% answered extremely satisfied. 5 out of 10 respondents or 50.00% answered very satisfied. Lastly, In terms of the Nutri-app Instructional Content Evaluation, "The content of the lesson represents an important curricular topic." 10 out of 10 respondents or 100.00% answered extremely satisfied.

6.4.2 Evaluation of Nutri-App in terms of Instructional Design

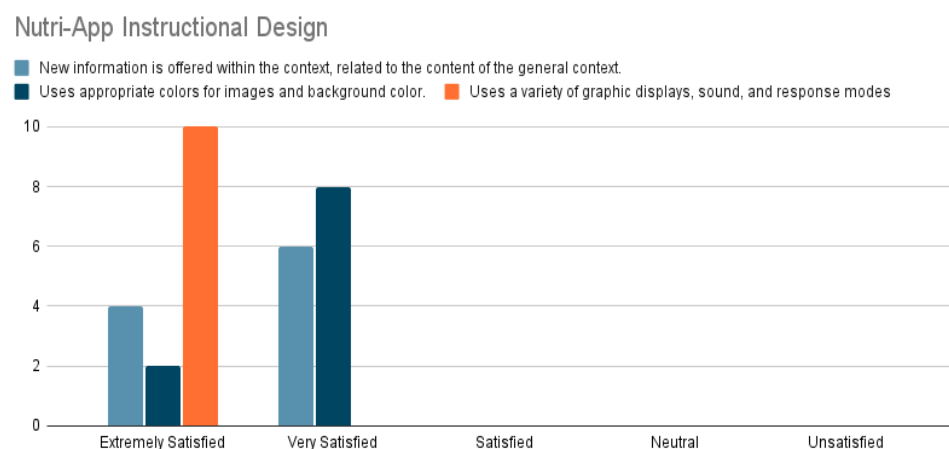


Figure 6.6 - Instructional Design Evaluation

In the figure above shows the result of the evaluation of the instructional design of the application. The respondents find that the design and color of the application are appropriate.

In terms of the Nutri-app Instructional Design Evaluation, "New information is offered within the context, related to the content of the general context." 4 out of 10 respondents or 40.00% answered extremely satisfied. 6 out of 10 respondents or 60.00% answered very satisfied.

Furthermore, in terms of the Nutri-app Instructional Design Evaluation, "Uses a variety of graphic displays, sound, and response modes." 10 out of 10 respondents or 100.00% answered extremely satisfied. Lastly, In terms of the Nutri-app Instructional Design Evaluation, "Uses appropriate colors for images and background color." 2 out of 10 respondents or 20.00% answered extremely satisfied. 8 out of 10 respondents or 80.00% answered very satisfied.

6.4.3 Evaluation of Nutri-App in terms of Management and Motivation

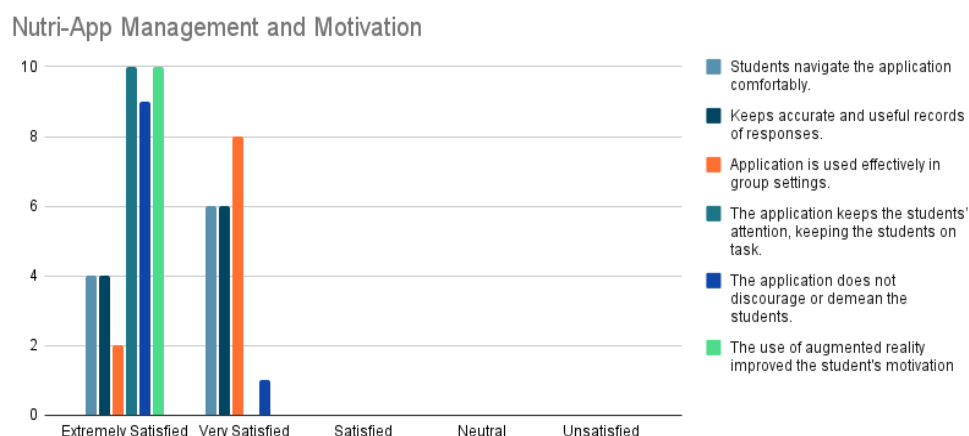


Figure 6.7 - Management and Motivation Evaluation

In the figure above shows the Management and Motivation of the application, assessed by teachers and student-teachers. It is evident that the use of augmented reality improved the students' motivation in learning nutritional concepts on this application.

In terms of the Nutri-app Management and Motivation Evaluation, "Students navigate the application comfortably." 4 out of 10 respondents or 40.00% answered extremely satisfied.

6 out of 10 respondents or 60.00% answered very satisfied. Moreover, in terms of the Nutri-app Management and Motivation Evaluation, "Keeps accurate and useful records of responses." 4 out of 10 respondents or 40.00% answered extremely satisfied. 6 out of 10 respondents or 60.00% answered very satisfied.

In terms of the Nutri-app Management and Motivation Evaluation, "Application is used effectively in group settings." 2 out of 10 respondents or 20.00% answered extremely satisfied. 8 out of 10 respondents or 80.00% answered very satisfied. In terms of the Nutri-app Management and Motivation Evaluation, "The application keeps the students' attention, keeping the student on task." 10 out of 10 respondents or 100.00% answered extremely satisfied. Furthermore, in terms of the Nutri-app Management and Motivation Evaluation, "The application does not discourage or demean the students." 9 out of 10 respondents or 90.00% answered extremely satisfied. 1 out of 10 respondents or 10.00% answered very satisfied. Lastly, in terms of the Nutri-app Management and Motivation Evaluation, "The use of augmented reality improved the students' motivation." 10 out of 10 respondents or 100.00% answered extremely satisfied.

6.4.4 Evaluation of Nutri-App in terms of Ease of Use

Nutri-App Ease of Use

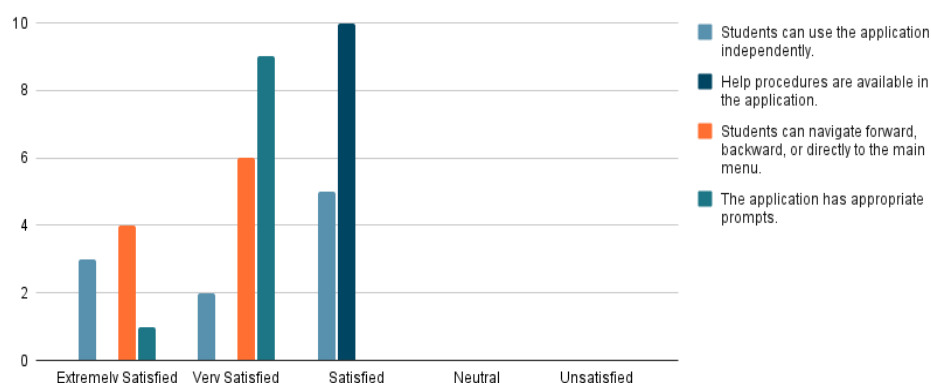


Figure 6.8 - Ease of Use Evaluation

In the figure above shows the Ease of Use of the application, assessed by teachers and student-teachers. Most of the students can use the application independently, students can navigate forward, backward, or directly to the menu.

In terms of the Nutri-app Ease of Use Evaluation, "Students can use the application independently." 3 out of 10 respondents or 30.00% answered extremely satisfied. 2 out of 10 respondents or 20.00% answered very satisfied. and 5 out of 10 respondents or 50.00% answered very satisfied.

Moreover, in terms of the Nutri-app Ease of Use Evaluation, "Help procedures are available in the application." 10 out of 10 respondents or 100.00% answered satisfied. In terms of the Nutri-app Ease of Use Evaluation, "Students can navigate forward, backward, or directly to the main menu." 4 out of 10 respondents or 40.00% answered extremely satisfied. 6 out of 10 respondents or 60.00% answered very satisfied. Lastly, in terms of the Nutri-app Ease of Use Evaluation, "The application has appropriate prompts." 1 out of 10 respondents or 10.00% answered extremely satisfied. 9 out of 10 respondents or 90.00% answered very satisfied.

6.4.5 Evaluation of Nutri-App in terms of Technical Aspects

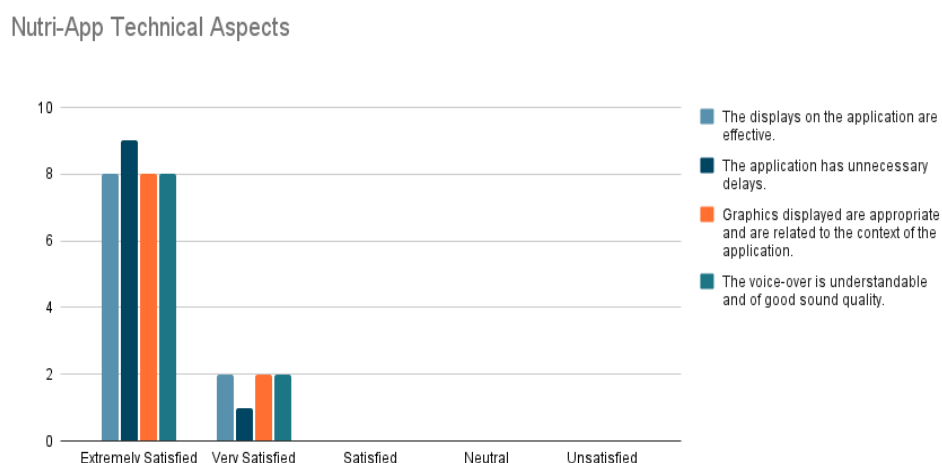


Figure 6.9 - Technical Evaluation

In the figure above shows the Instructional Design of the application, assessed by teachers and student-teachers. Most of the 10 respondents agreed that display on the application is effective. The graphics displayed are appropriate and related to the context of the application, and the voice-over of the application is of good quality.

In terms of the Nutri-app Technical Evaluation, "The displays on the application are effective." 8 out of 10 respondents or 80.00% answered extremely satisfied. 2 out of 10 respondents or 20.00% answered very satisfied.

In terms of the Nutri-app Technical Evaluation, "The has unnecessary displays." 9 out of 10 respondents or 90.00% answered extremely satisfied. 1 out of 10 respondents or 10.00% answered very satisfied. Moreover, In terms of the Nutri-app Technical Evaluation, "Graphics displayed are appropriate and are related to the context of the application." 8 out of 10 respondents or 80.00% answered extremely satisfied. 2 out of 10 respondents or 20.00% answered very satisfied. Lastly, in terms of the Nutri-app Technical Evaluation, "The voice-over is understandable and of good sound quality." 8 out of 10 respondents or 80.00% answered extremely satisfied. 2 out of 10 respondents or 20.00% answered very satisfied.

6.5 Summary of Findings

The study aimed to provide an alternative learning tool to help learners gain knowledge about nutritional concepts in a fun and interesting way. The development of an application that incorporates augmented reality led to positive responses from respondents, with most being extremely satisfied with the application's content and design.

Regarding the Nutri-app content, the respondents found the words easy to read, the lessons clear and easy to understand, and the voice-over clear and easy to understand. Additionally, they found the steps helpful, and they found the overall content of the application understandable. On the Nutri-app design and sounds evaluation, the respondents liked the sound effects, colors, and images on the application. However, some respondents were neutral about the buttons of the application, and some were neutral about the overall attractiveness of the application.

The Nutri-app engagement and interactivity evaluation, the respondents liked scanning the images that convert into augmented reality, found the games fun to play, and found the quiz game easy to play. However, some respondents found the countdown timer pressuring, and some were neutral about the engaging and interactive aspects of the application. On the other hand, the Nutri-app overall evaluation, the respondents liked learning from the application and earning rewards from it. However, some respondents found it took a long time to learn how to use the application, and some were neutral about sharing the application with classmates and friends.

The teachers and student-teachers also recognized the impact of the augmented reality integration on improving students' motivation to learn, as evidenced by the study's data. The study also revealed that while existing traditional learning methods still play a vital role in education, the use of modern technologies such as augmented reality and interactive games can significantly enhance the learning process. The engagement of e-learning, particularly with the current accessible technology, were also apparent in the study.

CHAPTER 7

CONCLUSIONS AND RECOMMENDATIONS

This chapter represents the conclusion and recommendation for the Nutri-App: Nutrition Learning Application Using Augmented Reality for Kids.

7.1 Conclusions

The use of the traditional way of learning nutrition concepts through textbooks on the elementary level has shown an opportunity to improve the imparting of knowledge to the learners. This research aimed to help the learners to have an alternative learning method by using modern technology such as the marker-based augmented reality with set of lessons and games to impart nutritional concepts to the learners. Two significant outcomes resulted on this study. First is the use of augmented reality to increase the engagement of using the application while learning nutritional concepts, second is the use of interactive games to increase the interactivity of the application.

The objectives of the application were met based on the summary of findings. First, it showed the success of integrating the learning concept of nutritional facts on the application based on the learning material of the DepEd. Based on the respondent's feedback, the content of the lessons on the application represents an important curricular topic. Second, the integration of marker-based augmented reality in the lesson mode is engaging. The users were entertained to use the augmented reality feature. Third, the use of color combination and appropriate design and images were pleasing on the end-user. Fourth, the integration of games on imparting nutritional concept were interactive and renders the users to use the application repeatedly. Lastly, the content, design, sound, usability, and technical features of the mobile app tested for usability were all good, but there is still room for development, particularly in the area of augmented reality and technical issues.

7.2 Recommendations

After evaluating the Nutri-app and considering the study's findings, the following recommendations are suggested:

- To improve the user experience, the Nutri-app should include clear and comprehensive instructions to help users navigate the application effectively.
- Including help or guide instructions in the Nutri-app can enhance the user experience by providing assistance for users to navigate the application more effortlessly.
- While it may not be feasible within the scope of this recommendation, incorporating augmented reality into the game mode in future updates could further enhance the learning experience.
- The application's error sound should be revised to provide a more informative and descriptive correction sound, enabling users to identify and correct their errors more quickly and easily.

By implementing these recommendations, the Nutri-app can be improved to provide a more effective and engaging learning tool for users.

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Tuli, N., Mantri, A. (2021). Evaluating Usability of Mobile-Based Augmented Reality Learning Environments for Early Childhood. *International Journal of Human-Computer Interaction*, 37(9).
<https://doi.org/10.1080/10447318.2020.184388>

APPENDICES

APPENDIX A: DATA GATHERING (Pictures)

Republic of the Philippines
National Education Policy Commission
DepEd Division Office - Marikina City
DepEd Marikina City

Name: Arvin B. B. Date: 10/10/22
School: DepEd Marikina City Grade & Section: 4-A

Evaluation for the Learners
This evaluation will be used to improve the effectiveness of the application.
The responses will be kept confidential and used for our study only.

Directions: Please fill out the following by using only one checkmark ✓ (1) per question.

	Very unsatisfied	Unsatisfied	Neutral	Satisfied	Very Satisfied
A. Content					
	☹	☹	😊	😊	😊
1. The words are easy to read.				✓	
2. The lessons are clear and easy to understand.					✓
3. The voice-over is clear and easy to understand.			✓		
4. The steps are helpful.				✓	
5. Overall, the content of the application is understandable.					✓

Design and Usability

	☹	☹	😊	😊	😊
1. I like the sound effects.		✓			✓
2. I like the colors of this application.				✓	
3. I like the images on this application.			✓		
4. I like the buttons of this application.				✓	
5. Overall, the application is attractive.					

C. Engagement and Interactivity

	☹	☹	😊	😊	😊
1. I like scanning the images that convert into augmented reality.				✓	
2. I found the quiz game easy to play.					✓
3. The games are fun to play.					✓
4. I find the countdown timer pressing.		✓			
5. Overall, the application is engaging and interactive.				✓	

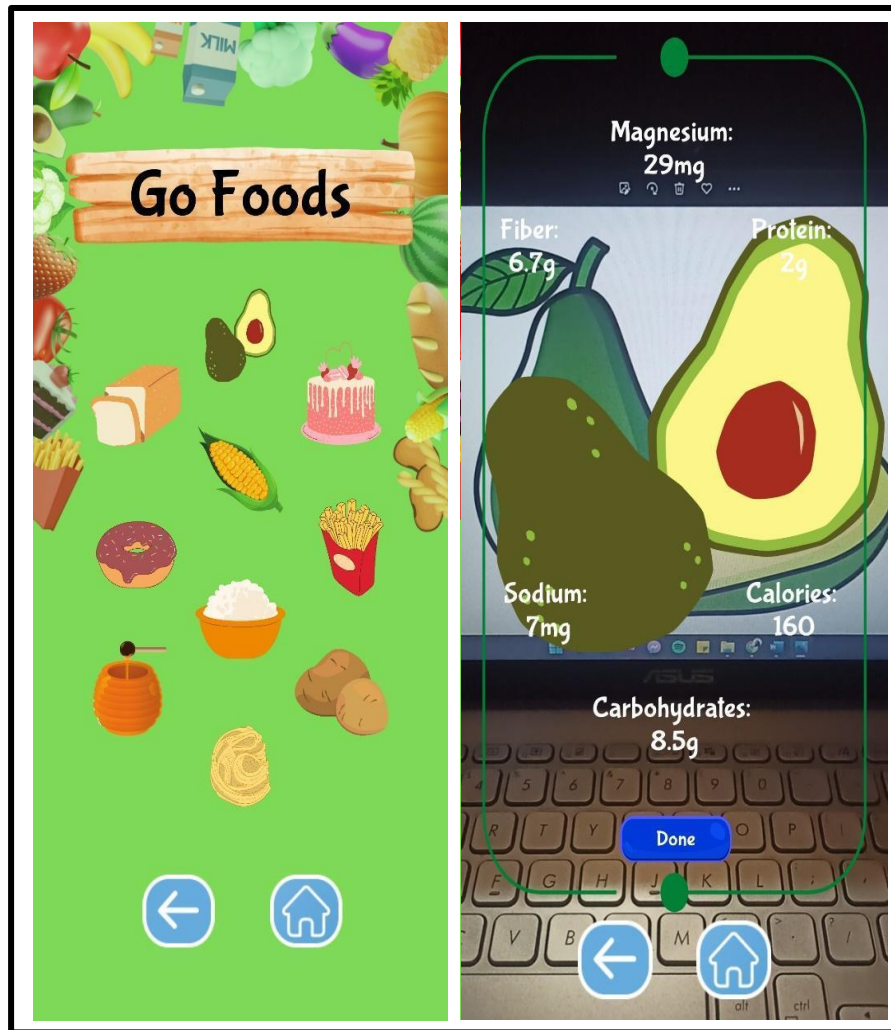
on Learning Application for Kids with Augmented Reality.

	☹	☹	😊	😊	😊
1. I like learning from this application.					✓
2. I like earning rewards from this application.			✓		
3. It took me a long time to learn how to use the application.					✓
4. I would share this application with my classmates and friends.			✓		

Suggestions.

APPENDIX B: Graphical User Interface (GUI)

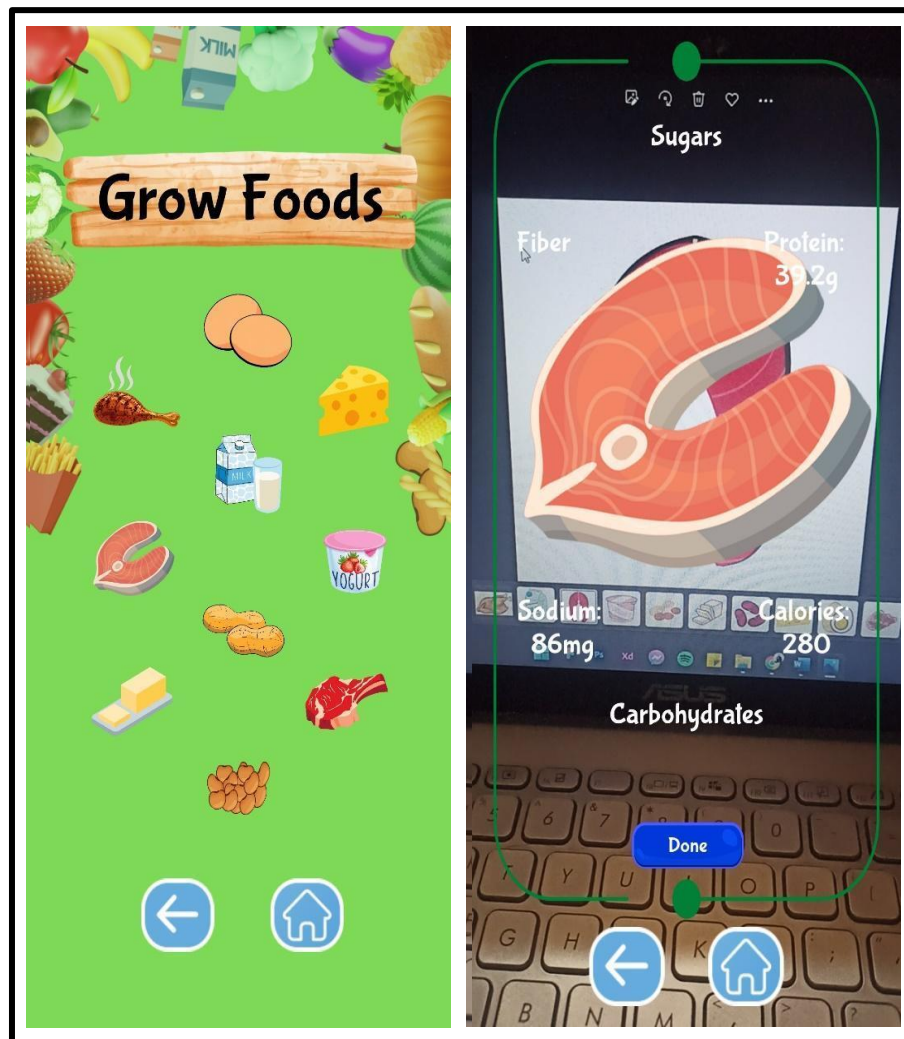
*GUI Main Screen and Lessons*



GUI Go Foods and Augmented Image



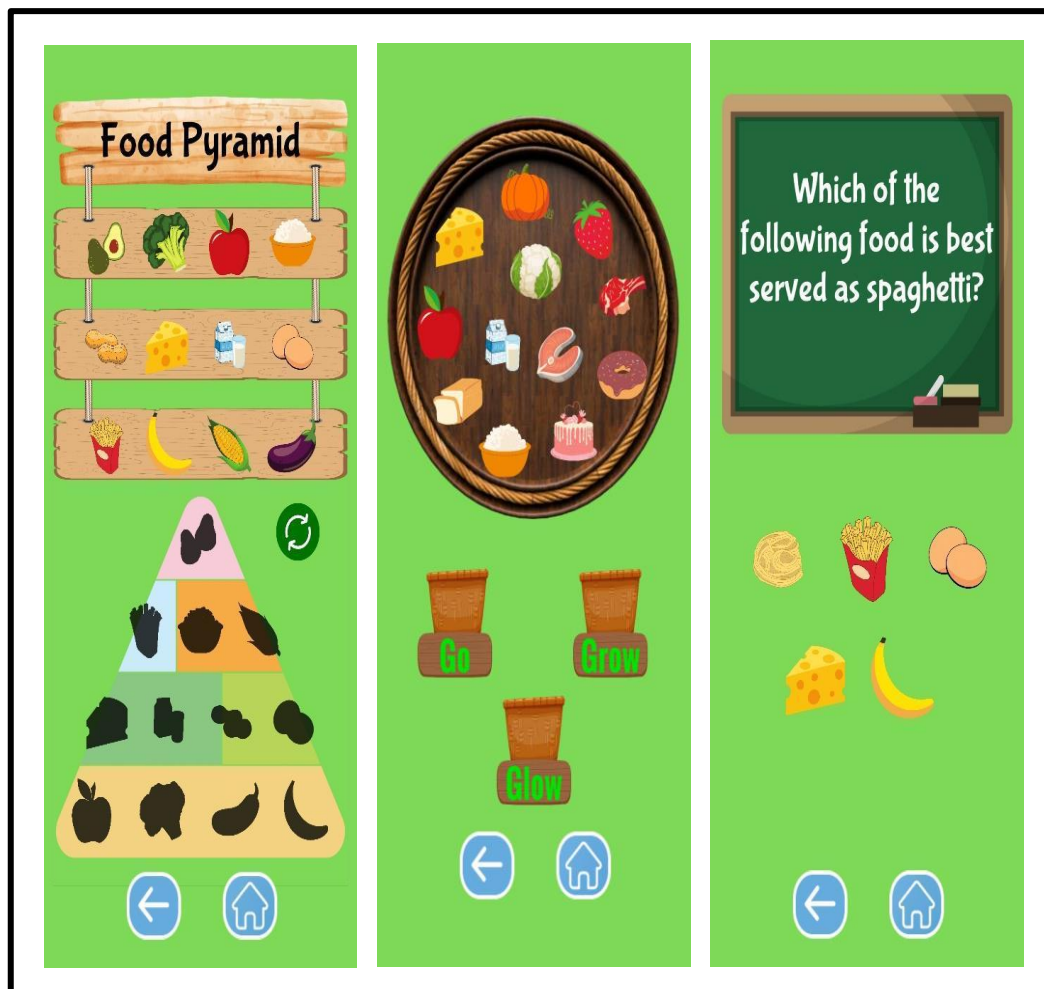
GUI Glow Foods and Augmented Image



GUI Grow Foods and Augmented Image



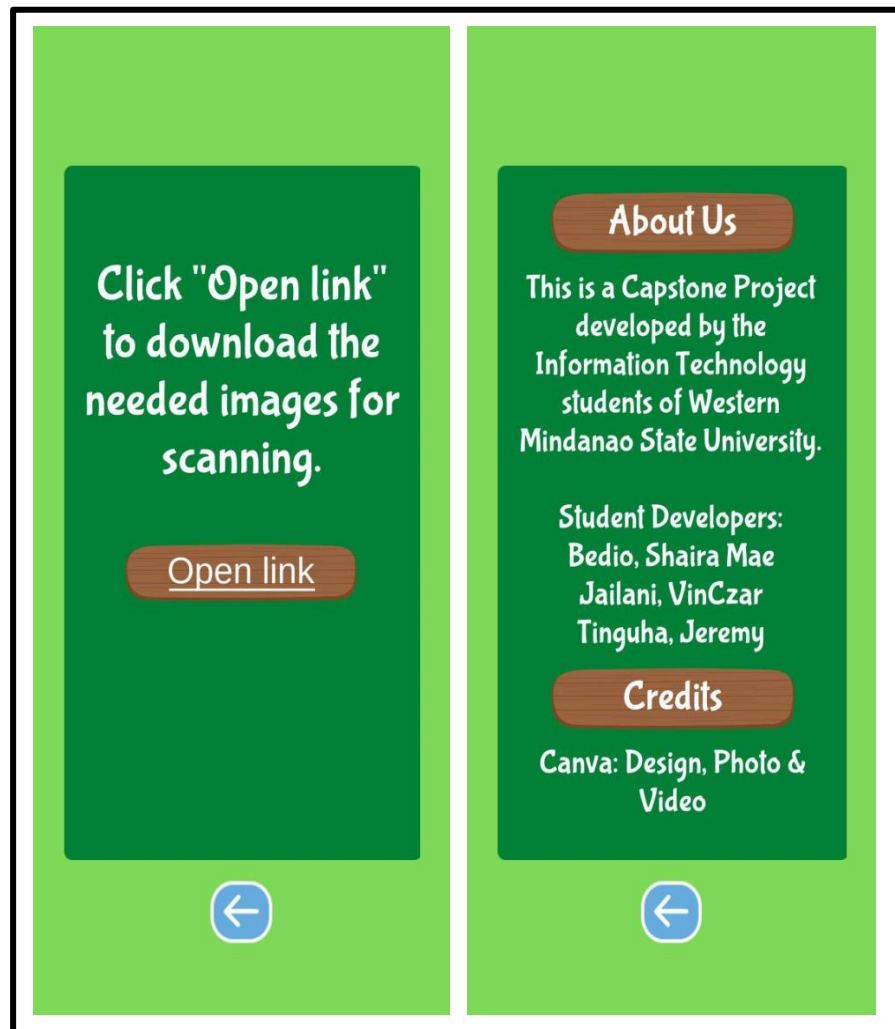
GUI Nutri-app Games



GUI Challenges and Quiz Reset

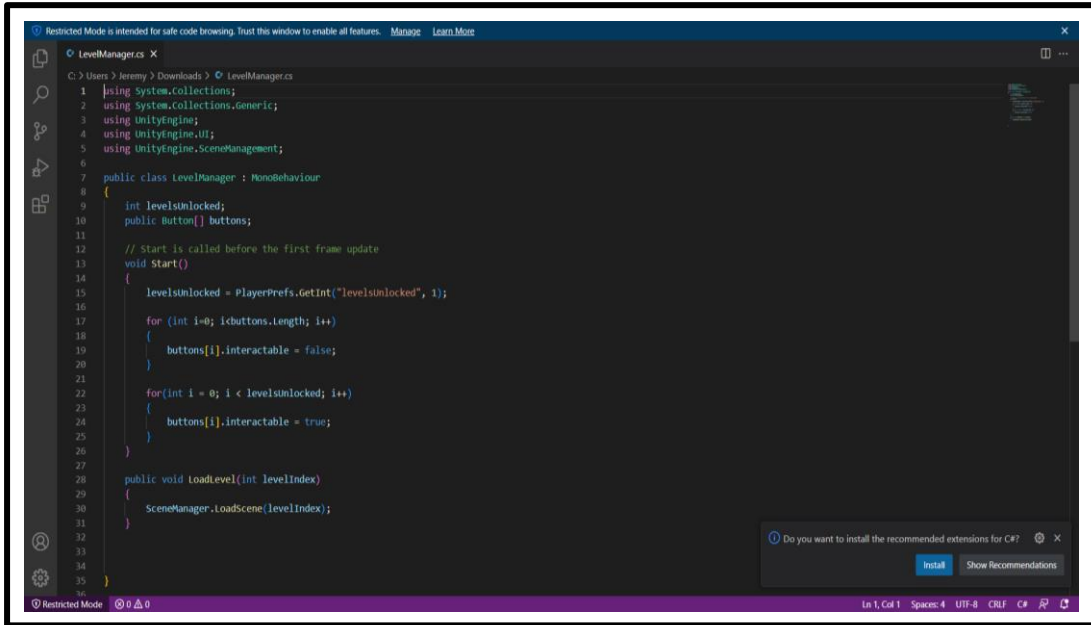


GUI Collection of Tokens and Food Pyramid Lesson

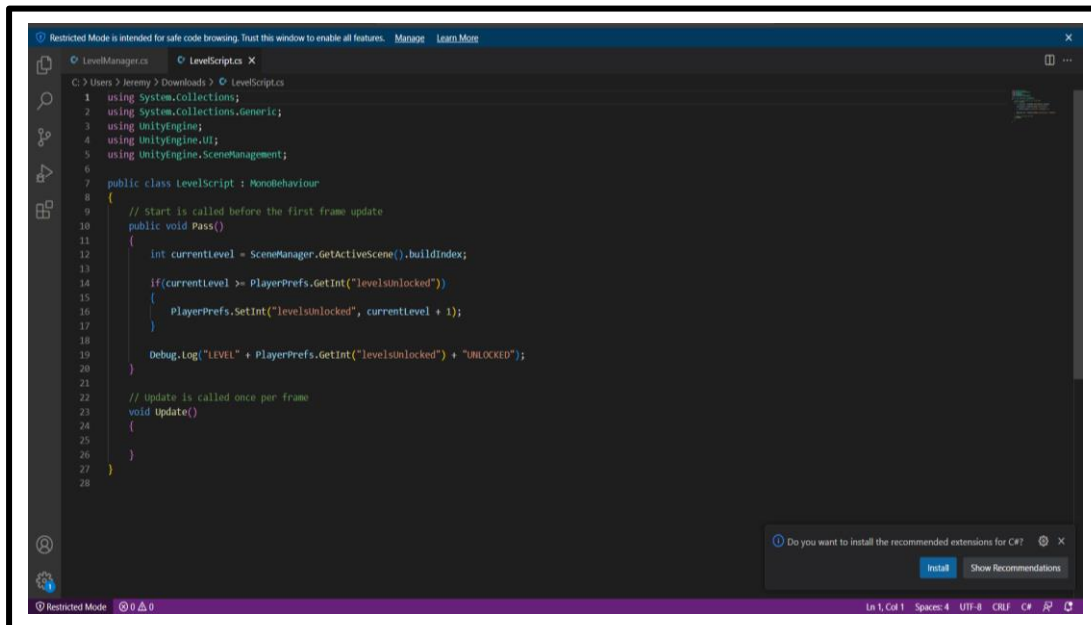


GUI Downloadable Link and About Us

Appendix C: Source Code



```
1 using System.Collections;
2 using System.Collections.Generic;
3 using UnityEngine;
4 using UnityEngine.UI;
5 using UnityEngine.SceneManagement;
6
7 public class LevelManager : MonoBehaviour
8 {
9     int levelsunlocked;
10    public Button[] buttons;
11
12    // Start is called before the first frame update
13    void Start()
14    {
15        levelsunlocked = PlayerPrefs.GetInt("levelsunlocked", 1);
16
17        for (int i = 0; i < buttons.Length; i++)
18        {
19            buttons[i].interactable = false;
20        }
21
22        for(int i = 0; i < levelsunlocked; i++)
23        {
24            buttons[i].interactable = true;
25        }
26    }
27
28    public void loadLevel(int levelIndex)
29    {
30        SceneManager.LoadScene(levelIndex);
31    }
32
33
34
35
36 }
```



```
1 using System.Collections;
2 using System.Collections.Generic;
3 using UnityEngine;
4 using UnityEngine.UI;
5 using UnityEngine.SceneManagement;
6
7 public class LevelScript : MonoBehaviour
8 {
9     // Start is called before the first frame update
10    public void Pass()
11    {
12        int currentLevel = SceneManager.GetActiveScene().buildIndex;
13
14        if(currentLevel >= PlayerPrefs.GetInt("levelsunlocked"))
15        {
16            PlayerPrefs.SetInt("levelsunlocked", currentLevel + 1);
17        }
18
19        Debug.Log("LEVEL" + PlayerPrefs.GetInt("levelsunlocked") + "UNLOCKED");
20    }
21
22    // Update is called once per frame
23    void Update()
24    {
25    }
26
27
28 }
```


Restricted Mode is intended for safe code browsing. Trust this window to enable all features. [Manage](#) [Learn More](#)

LevelManager.cs LevelScripts.cs LevelManager (1).cs LevelManager.cs.meta Manager.cs X

C:\Users\Jeremy\Downloads> Manager.cs

```

91
92
93 public void DropAvocado()
94 {
95     float Distance = Vector3.Distance(avocado.transform.position, avocadoBlack.transform.position);
96     if (Distance < 50)
97     {
98         avocado.transform.position = avocadoBlack.transform.position;
99         source.clip = correct[Random.Range(0, correct.Length)];
100         source.Play();
101         avocadoCorrect = true;
102     }
103     else
104     {
105         avocado.transform.position = avocadoInitialPos;
106         source.clip = incorrect;
107         source.Play();
108     }
109 }
110
111 public void DropFries()
112 {
113     float Distance = Vector3.Distance(fries.transform.position, friesBlack.transform.position);
114     if (Distance < 50)
115     {
116         fries.transform.position = friesBlack.transform.position;
117         source.clip = correct[Random.Range(0, correct.Length)];
118         source.Play();
119         friesCorrect = true;
120     }
121     else
122     {
123         fries.transform.position = friesInitialPos;
124         source.clip = incorrect;
125         source.Play();
126     }
127 }

```

The Marketplace has extensions that can help with '.meta' files. [Search Marketplace](#) [Don't Show Again for '.meta' files](#)

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LevelManager.cs LevelScripts.cs LevelManager (1).cs LevelManager.cs.meta Manager.cs X

C:\Users\Jeremy\Downloads> Manager.cs

```

4 public class Manager : MonoBehaviour
5 {
6
7
8     public GameObject avocado, fries, corn, cheese, milk, peanut, egg, apple, broccoli, eggplant, banana, rice, avocadoBlack, friesBlack, cornBlack, cheeseBlack, milkBlack;
9
10     Vector2 avocadoInitialPos, friesInitialPos, cornInitialPos, cheeseInitialPos, milkInitialPos, peanutInitialPos, eggInitialPos, appleInitialPos, broccoliInitialPos, eggplantInitialPos, bananaInitialPos, riceInitialPos;
11
12     public static bool locked;
13
14     public AudioSource source;
15     public AudioClip[] correct;
16     public AudioClip incorrect;
17
18     bool avocadoCorrect, friesCorrect, cornCorrect, cheeseCorrect, milkCorrect, peanutCorrect, eggCorrect, appleCorrect, broccoliCorrect, eggplantCorrect, bananaCorrect, riceCorrect;
19
20     [SerializeField] private GameObject winText;
21
22     void Start()
23     {
24
25         avocadoInitialPos = avocado.transform.position;
26         friesInitialPos = fries.transform.position;
27         cornInitialPos = corn.transform.position;
28         cheeseInitialPos = cheese.transform.position;
29         milkInitialPos = milk.transform.position;
30         peanutInitialPos = peanut.transform.position;
31         eggInitialPos = egg.transform.position;
32         appleInitialPos = apple.transform.position;
33         broccoliInitialPos = broccoli.transform.position;
34         eggplantInitialPos = eggplant.transform.position;
35         bananaInitialPos = banana.transform.position;
36         riceInitialPos = rice.transform.position;
37
38         winText.SetActive(false);
39     }
40 }

```

The Marketplace has extensions that can help with '.meta' files. [Search Marketplace](#) [Don't Show Again for '.meta' files](#)

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Appendix D

Table 5.1 Unit Testing

ID	Test Case	Expected Result	Pass/ Failed
1	User starts the application.	The user is on the loading screen and then the main menu.	PASS
2	User selects the lesson mode.	The user is in lesson mode with 4 options.	PASS
3	User selects the glow foods category.	The user is in the glow foods category, with 10 various food to scan.	PASS
4	User selects the go foods category.	The user is in the go foods category, with 10 various food to scan.	PASS
5	User selects the grow foods category.	The user is in the grow foods category, with 10 various food to scan.	PASS
6	User selects the food pyramid.	The user is the lesson mode of the food pyramid.	PASS
7	User selects the challenges mode.	The user is in subsequent games once passed/unlocked.	PASS
8	User selects the first challenge. (Fill-out the Food Pyramid)	The user is in the fill-out-the-food pyramid challenge.	PASS
9	User selects the second challenge. (Fill-the-Basket)	The user is in the fill-the-basket challenge.	PASS

Table 5.2 UI Testing

Characteristics	Description	Pass/Failed
Consistency	Consistent navigation of the application.	PASS
Functionality	Errors/Bugs free.	PASS
Usability	Ease-of-use.	PASS
Typography	Texts are legible.	PASS
Color and Design	Appropriate for the target users.	PASS

Table 5.2 User Acceptance Testing

ID	ACCEPTANCE REQUIREMENTS	ACCEPT/REJECT
1	The application must properly operate althroughout.	ACCEPT
2	The application must help the target users to learn the basic of nutritional concepts from DepEd materials.	ACCEPT
3	The application must be interactive.	ACCEPT
4	The marker-based augmented reality must be integrated within the lessons module.	ACCEPT
5	In challenges mode, the users must be able to unlock by succeeding the current level.	ACCEPT

Appendix E: Evaluation Form

For Teachers:

	Republic of The Philippines Western Mindanao State University COLLEGE OF COMPUTING STUDIES DEPARTMENT OF INFORMATION TECHNOLOGY	
---	---	---

Name (optional): _____ Date: _____
 School: _____ Age: _____

Evaluation of the Teachers

This evaluation will be used to determine the effectiveness of the application.
 The responses will be kept confidential and used for our study only.

Directions: Please respond to the following question by placing a check mark (✓) in the box that corresponds to your answer.

1 - Unsatisfied	2 - Neutral	3 - Satisfied	4 - Very Satisfied	5 - Extremely Satisfied
-----------------	-------------	---------------	--------------------	-------------------------

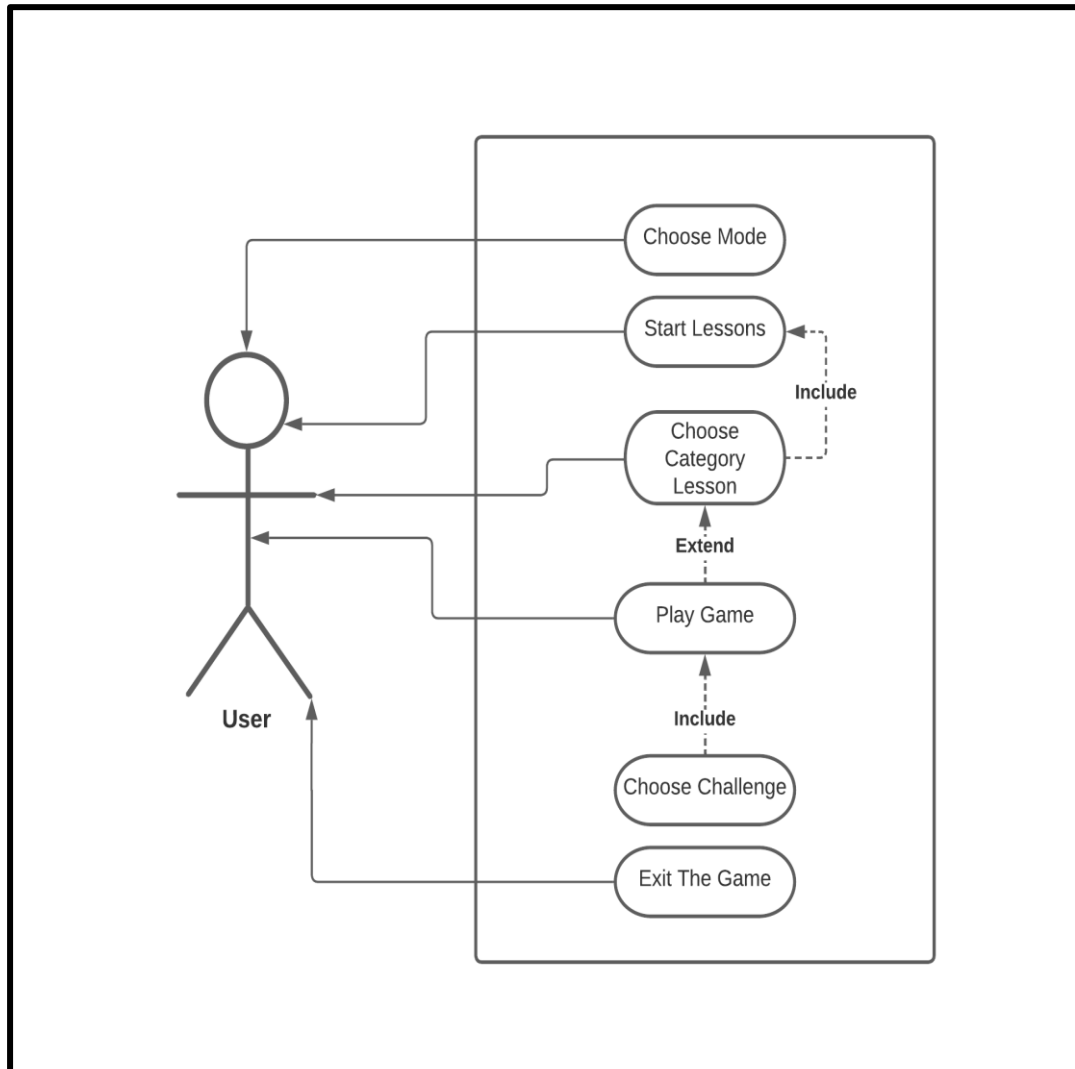
A. Instructional Content					
	1	2	3	4	5
1. The objectives of the application are well-defined.					
2. Content is sequenced (tasked analyzed) appropriately.					
3. Contents of the games are appropriate for lesson objectives.					
4. The level of difficulty is suitable for the target learners. (e.g. content, vocabulary, readability.)					
5. For frequent users, the application offers variations of features in the application.					
6. The content of the lesson represents an important curricular topic.					

B. Instructional Design					
	1	2	3	4	5
1. New information is offered within the context, related to the content of the general context.					
2. Uses appropriate colors for images and background color.					
3. Uses a variety of graphic displays, sound, and response modes appropriately.					
C. Management and Motivation					
	1	2	3	4	5
1. Students navigate the application comfortably.					
2. Keeps accurate and useful records of responses.					
3. Application is used effectively in group settings.					
4. The application keeps the students' attention, keeping the students on task.					
5. The application does not discourage or demean the students.					
6. The use of augmented reality improved the student's motivation in learning and use the application.					
D. Ease of Use					
	1	2	3	4	5
1. Students can use the application independently.					
2. Help procedures are available in the application.					
3. Students can navigate forward, backward, or directly to the main menu.					
4. The application has appropriate prompts.					
E. Technical Aspects					
	1	2	3	4	5
1. The displays on the application are effective.					
2. The application has unnecessary delays.					
3. Graphics displayed are appropriate and are related to the context of the application.					
4. The voice-over is understandable and of good sound quality.					
Write your recommendations here.					

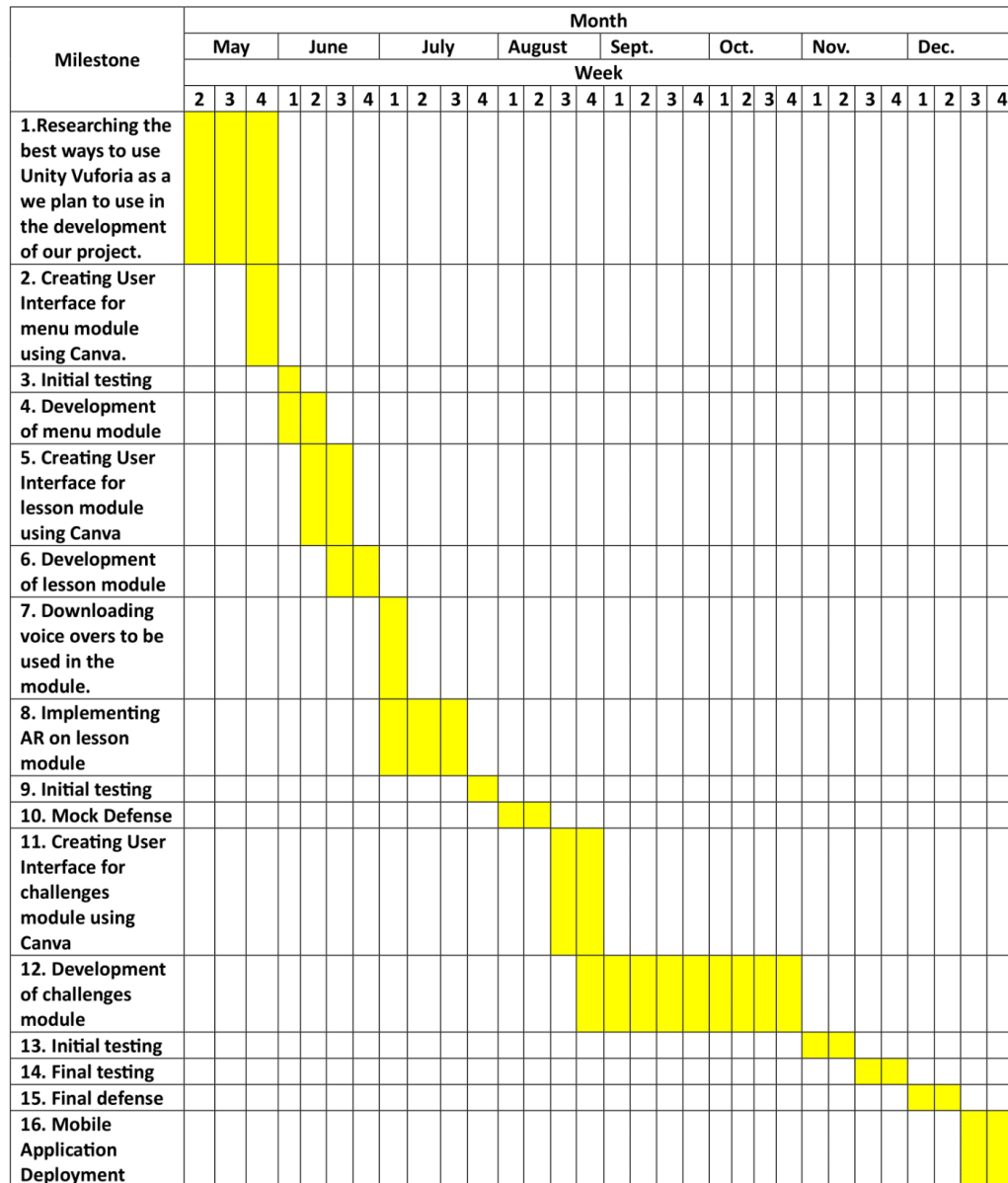
For Learners:

	Republic of The Philippines Western Mindanao State University COLLEGE OF COMPUTING STUDIES DEPARTMENT OF INFORMATION TECHNOLOGY				
Name (optional): _____ Date: _____ School: _____ Grade & Section: _____					
Evaluation for the Learners This evaluation will be used to determine the effectiveness of the application. The responses will be kept confidential and used for our study only.					
Directions: Please fill out the following by using only one checkmark ✓ (1) per question.					
	Very unsatisfied	Unsatisfied	Neutral	Satisfied	Very Satisfied
A. Content					
					
1. The words are easy to read.					
2. The lessons are clear and easy to understand.					
3. The voice-over is clear and easy to understand.					
4. The steps are helpful.					
5. Overall, the content of the application is understandable.					
B. Design and Sounds					
					
1. I like the sound effects.					
2. I like the colors of this application.					
3. I like the images on this application.					
4. I like the buttons of this application.					
5. Overall, the application is attractive.					
C. Engagement and Interactivity					
					
1. I like scanning the images that convert into augmented reality.					
2. I found the quiz game easy to play.					
3. The games are fun to play.					
4. I find the countdown timer pressuring.					
5. Overall, the application is engaging and interactive.					

Appendix F: Use Case Diagram



Appendix G: Gantt Chart



Curriculum Vitae

Project Manager



VINCZAR R. JAILANI

Pasay Drive, San Roque Zamboanga City
(+63) 935-617-5109
vinczarj@gmail.com

PERSONAL DETAILS

Nationality: Filipino

Date of Birth: July 22, 2000

Place of Birth: Puericulture Hospital, Zamboanga City

WORK EXPERIENCE

Summer Work Job

Zamboanga City, Philippines

- Assisting local officials on their clerical works.
- Conducted surveys tasked by the Philippine Statistics Authority.

EDUCATIONAL BACKGROUND

Tertiary (2018-2023)

Bachelor of Science in Information Technology.
Western Mindanao State University.
Normal Road, Baliwasan Zamboanga City.

Secondary (2012-2018)

Science Technology Engineering and Mathematics.
Pilar College of Zamboanga City Incorporated.
San Jose Rd, 7000 Zamboanga City.
(2016-2018)

Zamboanga National High School West.
Normal Road, Baliwasan Zamboanga City.
(2012-2016)

TRAINING/SEMINAR/WORKSHOP

First Crimson Young Leader's Congress 2020

Western Mindanao State University – University Student Council
Dr. Juanito Bruno Gymnasium, Western Mindanao State University
November 24-26, 2020

Young Leaders for Resilience Program: Enterprise Design Thinking Workshop.

IBM Philippines, National Resilience Council and ARISE Philippines.
Centro Latino Paseo Del Mar Zamboanga City.
September 06-07, 2019.

Work immersion.

Pag-IBIG Fund. Normal Road, Baliwasan Zamboanga City.
December, 2017

ACHIEVEMENTS

Awards:

- Oral Expression Awardee, 2018

Affiliations:

Editor-in-Chief – Venom Publication (College Student Publication)

2021 – Present

Chief Operations Officer and Internal Vice-President

Google Developer Student Club Crimson

2021 – Present

Senator – Western Mindanao State University Student Council

2020 – Present

Associate Editor – Venom Publication (College Student Publication)

2019 – 2020

Councilor - Institute of Computer Studies Student Council

2019 – 2020

Representative – Philippine Society of Information Technology Students

2018 – 2019

SKILLS

- Administration (Experienced in administrating several organizations)
- Management (Organizing events and Organizations)
- Language Proficiency (Proficient in English)
- Charismatic and confident public speaker.
- Excellent communication skills.

REFERENCES

Ms. Yusmita Miranda, LPT.

Teacher I

Sinunuc National High School

(+63)945-2800-214

yusmitamiranda1209@gmail.com

Mr. Ernel Miranda, LPT.

Teacher I

Sta. Maria Elementary School

(+63)995-8423-745

ernelmiranda@gmail.com

Ms. Charlene Tubaing, RSW.

Financial Manager

W. Hydrocolloids Inc.

(+63)975-0322-710

chalenet14@gmail.com

Lead Developer



JEREMY TINGUHA

ADDRESS

Atis Drive, Baliwasan
Chico Zamboanga City,
Philippines 7000

CONTACT

(+63) 9052433491
jemtinguha04@gmail.com

PERSONAL DETAILS

Date of Birth: July 04, 2000
Place of Birth:
Ubit, Lamitan City, Basilan
Nationality: Filipino



EDUCATIONAL BACKGROUND

TERTIARY

BACHELOR OF SCIENCE IN INFORMATION TECHNOLOGY
Western Mindanao State University | 2028 - 2022
Baliwasan Normal Rd., Zamboanga City

SECONDARY

SENIOR HIGH
TECHNICAL-VOCATIONAL-LIVELIHOOD ICT
Zamboanga del Sur National High School | 2016 - 2018
Sta, Maria District, Pagadian City, Zamboanga del Sur

JUNIOR HIGH
Basilan National High School | 2012 - 2016
PEO Rd, Isabela City, Basilan



TRAINING/SEMINAR

GRAPHIC DESIGNING
Shaw Academy(Webinar) | August 2020

WORK IMMERSION
Metro Grapix | March 2018
Rizal Avenue, pagadian City



ACHIEVEMENTS

BEST SOFTWARE ENGINEERING PROJECT
Western Mindanao State University
College of Computing Studies



SKILLS

- Computer Literate
- Graphic Designing
- Good communication skill
- Strategic Thinking
- Creative
- Sense of Leadership
- Teamwork
- Flexible



CHARACTER REFERENCE

MS. DENNY ALOVERA
Email Support | Collective Solution Philippines
Coach - Crimson Hawks All Stars
09553315919

ENGR. RAFFI MULOC
Software Developer | Orchid
Coach - Crimson Hawks All Stars
09569303690 | raffibucog@gmail.com

ENGR. ABDLHAFIZ ABDURASID
National Commission on Muslim Filipinos Region IX-B |
Development Management Officer/Secretary
Coach - Crimson Hawks All Stars
09176524986 | abdlhafiz.abdurasid@gmail.com

Quality Assurance Analyst



Shairamae Bedio

☎ (+63) 906 868 1269

✉ bedioshairamae@gmail.com

📍 Zamboanga City

ABOUT ME

Hard-working, a fast learner, creative and proactive. Eager to keep growing, developing new skills and abilities.

SKILLS

- Communication
- Problem Solving
- Positivity
- Adaptability
- Motivation
- Able to work well under pressure
- Microsoft Word
- Microsoft Excel
- Microsoft Power Point
- Healthcare
- First aid

LANGUAGES

English
Tagalog
Chavacano
Bisaya

REFERENCES

Daniel Angelito G Hernandez

Junior Teacher 1
Claret School of Zamboanga City
danielhernz25@gmail.com
daniel.hernandez@cszc.edu.ph
(+63)-916-139-4807

Marjorie Rojas

Institute of Computer Studies Instructor
Western Mindanao State University
Marjorierojas@wmsu.edu.ph
(+63)-917-710-3254

PERSONAL DETAILS

Age : 22
Gender : Female
Civil Status : Single
Nationality : Filipino
Date of Birth : January 12, 2000
Religion : Roman Catholic

EDUCATION

Western Mindanao State University

Bachelor of Science in Information Technology
Normal Road, Baliwasan, Zamboanga City
2018 - current

Claret School of Zamboanga City

Grade school & High School
Ruste Drive, San Jose Road, Zamboanga City
2006/2018

TRAINING/SEMINAR/WORKSHOP

Work Immersion

Ciudad Medical de Zamboanga
Mayor, Vitaliano Agan Ave, Zamboanga, 7000
Zamboanga del Sur
50 hrs
2018

Career Fest

Claret School of Zamboanga City
Ruste drive, San Jose Rd, Zamboanga City
2018

Standard First Aid and Basic Life Support

Claret School of Zamboanga City
Ruste drive, San Jose Rd, Zamboanga City
2017